

Enhancing Online Safety: A Text Analysis Approach For Cyberbullying Detection In Social Media Tweets

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Abstract

In the modern digital age, cyberbullying has emerged as a concerning issue which threatens the safety of social media users. This research aims to tackle the overwhelming challenge of detecting cyberbullying incidents within millions of tweets. Motivated by the need to enhance online safety and minimize the negative psychological effects of cyberbullying, the project gathered massive social media tweet from several datasets, adopted CRISP-DM methodology and natural language processing (NLP) techniques to sift through massive datasets of tweets, creating cyberbullying detection models. The first part of this project ends with data preprocessing, prepared the dataset for applying feature extraction and model building. The significance of this project lies in its potential to address and mitigate one of the more insidious aspects of digital communication – cyberbullying, underscoring a pivotal step towards fostering safer digital environments. This initiative aligns with the United Nations' Sustainable Development Goal (SDG) 16, promoting peace, justice, and strong institutions by attempting to counteract online harm.

Keywords: Cyberbullying Detection, Social Media Tweets, Text Analysis, Natural Language Processing (NLP), Text Classification, Feature Extraction

SDG Goal 16: Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels

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CHAPTER 1: INTRODUCTION

1.1 Introduction

The widespread use of social media platforms in the digital age has fundamentally transformed the way we connect to our daily lives, providing tremendous chances for communication, connection, and community-building. These platforms have brought people across the world closer together, fostering a new sense of global interconnectedness like never before. However, the internet has become an inseparable element of human functioning, bringing undeniable benefits but also a source of many threats. (Szumiec, 2023) The early days of the internet saw concerns primarily focused on privacy and data security, but as the internet became more social with platforms like Facebook, Twitter, and Instagram, the focus shifted towards user behaviour and safety.

Cyberbullying is defined as the use of electronic communication to intimidate someone, usually by sending messages of a threatening. It includes various forms such as harassment, flaming, denigration, impersonation, outing, trickery, exclusion, and cyberstalking. (Tintori et al., 2023) Each type of cyberbullying has its unique characteristics and impacts, making it a complex phenomenon to address. This form of harassment, carried out through digital platforms, has profound implications, affecting the mental and emotional well-being of individuals, particularly the youth. (Chan et al., 2021) The repercussions of cyberbullying extend beyond the immediate victim, casting a shadow on the overall health and safety of online communities. (Garett et al., 2016)

To combat this growing menace, the development of cyberbullying detection systems has emerged as a promising solution. (Yi & Zubiaga, 2023) These systems aim to proactively identify instances of online bullying, thereby enabling timely interventions and creating safer online environments. These systems, leveraging advanced technologies, aim to identify and mitigate instances of harmful content. By analysing content in real-time, they offer the potential to intervene before harm can escalate, serving as a digital guardian against malicious intent. (Tahmasbi & Fuchsberger, 2019) However, the dynamic and diverse nature of social media presents a myriad of challenges for these detection systems. Tweets, posts, and comments are often laden with sarcasm, cultural nuances, evolving slang, and contextual ambiguities, making it difficult for automated systems to consistently discern intent. (Aziz et al., 2022) Moreover,

the sheer volume of content generated on popular platforms adds to the complexity, demanding solutions that are not only accurate but also scalable.

Thus, the ever-changing landscape of social media requires that these systems be adaptable and continually updated. Continuous research, development, and refinement are imperative to enhance the effectiveness and reliability of these detection systems, ensuring that social media remains a space of positive interaction and genuine connection.

1.2 Problem Background

Cyberbullying detection systems are tools that help identify bullying behaviour on social media. The systems use data analysis to sift through large amounts of online conversations and spot harmful content. The problems of cyberbullying detection systems can be discussed in terms of both social media and data analytics:

1. **Rising Incidence of Cyberbullying**

Cyberbullying is still prevalent on social media platforms today, harming people and negatively affecting how people use the internet. The prevalence of cyberbullying events has rapidly expanded with the increased use of social media, posing a significant threat to online safety and the emotional well-being of users. (Huong, 2023) As this problem worsens, there is an urgent need for the creation of efficient tools and tactics for spotting and preventing cyberbullying in social network tweets. (Qiu et al., 2022) Addressing this issue is crucial for preventing problems like depression, suicidal thoughts, and other mental health difficulties that can arise from the unpleasant experience of online harassment in addition to improving online safety.

2. **Complexity of Natural Language**

The unique nature of tweets, including their brevity and the use of informal language constructs, poses a significant challenge for text analysis. Tweets often contain emojis, abbreviations and hashtags, which are not typically found in formal writing. (Magumba & Nabende, 2021) These elements can convey nuanced meanings that are not easily captured by algorithms designed for more traditional text. Social media content often contains sarcasm, humour, cultural references and local slang that make the content highly ambiguous, can be

misinterpreted by cyberbullying detection system. This is further complicated by the limited context available in isolated posts or comments, making it challenging to discern the true intent behind a statement. Furthermore, the vast diversity of users brings about cultural and regional differences. (Görzig et al., 2017) What might be perceived as offensive or bullying in one culture or region could be deemed acceptable or even complimentary in another, adding another layer of complexity to the detection process. While most of the cyberbullying research focuses on English-language content, social media's global spread needs the creation of models capable of detecting cyberbullying across multiple languages and cultural contexts.

3. Detection Modelling Challenges

The language used on Twitter is not only unique but also highly dynamic. It reflects the latest cultural trends, news events, and social movements, often leading to the rapid creation and adoption of new lexicon. (Singh et al., 2022) This evolving nature of language requires continuous updates to the lexicon and algorithms used in text analysis to maintain accuracy. Most of the cyberbullying detection systems are unable to fully recognize the complex patterns of cyberbullying behaviour. (Raj et al., 2022) Moreover, these systems often suffer from a lack of ongoing training, failing to adapt to the ever-changing landscape of online communication. This leads to a critical gap in the ability to provide a safe and supportive online environment. This necessitates a more dynamic and responsive approach to cyberbullying detection that can keep pace with the changing landscape of social media interactions.

4. Data Integrity and Preparation

The effectiveness of cyberbullying detection system heavily relies on the quality and representativeness of the data used for training and validation. Inconsistent, incomplete, or noisy data can significantly skew the training of detection algorithms, leading to unreliable results. Furthermore, data must be appropriately tagged and annotated, which is a time-consuming procedure that also necessitates an in-depth knowledge of language, context, and cultural differences. (Dewani et al., 2021) The complexity of cyberbullying is compounded by the fact that what constitutes cyberbullying can vary significantly across contexts and people.

1.3 Project Aim

The aim of this project is to develop a cyberbullying detection model that utilizes text analysis techniques to identify and mitigate cyberbullying in social media tweets effectively.

1.4 Objectives

The objectives of this project are:

1. To build cyberbullying detection models that integrate text analysis techniques to identify cyberbullying in social media tweets.
2. To categorize cyberbullying content into distinct types, including age, gender, racism, religion, aggression and toxicity, for precise identification and response.
3. To evaluate the accuracy and performances of cyberbullying detection models.
4. To deploy the best cyberbullying detection model to identify contents of tweets.

1.5 Scope

1.5.1 Dataset

The data collection for this project involved gathering several datasets comprising information from various sources related to the detection of cyberbullying collected from Kaggle.com. These datasets included comments from popular social media platforms such as Twitter and YouTube. The dataset consisted of text data only, each labelled as either indicative of cyberbullying or not. Some of them also defined what constitutes cyberbullying, including age-based bullying, cyberbullying based on race (racism), gender (sexism), religious harassment, and general aggression and toxicity.

1.5.2 Deliverables

Developing a text classification based cyberbullying detection solution is a multi-step process that begins with data collection and ends with machine learning classification.

1. Data Understanding & Data Preprocessing

Understanding the unstructured text data can be an iterative and exploratory process. Some data understanding techniques such as getting basic information of the datasets will be conducted for helping the researcher to do further steps on the datasets. Data preprocessing includes cleaning (remove symbols and stop words), normalization (lowercasing), stemming and tokenization to ensure the data can be processed by the next step.

2. Apply Text Mining and Natural Language Processing Techniques

After preprocessing, text mining and natural language processing techniques are applied. This includes feature extraction tasks like Bag of Words, TF-DIF and word embeddings. These techniques help in understanding the context, themes, sentiments, and key entities present in the text data. Feature extraction and selection is emphasized in this stage, where specific attributes of the text data are identified and extracted.

3. Model Training

The dataset will be split into training, validation and testing sets before the model building. Training data shall be used for the development of models, validation data will help to determine their efficacy and test data are taken into account when evaluating model performance. In this phase, classification of text data according to categories such as bullying and nonbullying is developed and trained by machine learning models. The machine-based algorithms, like Nave Bayes and Support Machines Vector (SVM) can be used for model building.

4. Model Evaluation

Evaluating the models are crucial to assess their effectiveness. The common method to evaluate the models is understanding the strengths and weaknesses of the model using a confusion matrix. (Desai et al., 2021) This matrix provides a clear breakdown of the model and predictions, revealing true positives, true negatives, false positives and false negatives. Another method that can be used to evaluate the model is the F1 score which measures the accuracy of

the trained models. (Evangeline et al., 2021) There are other ways to evaluate the models such as generating performance metrics using Lazy Predict, Precision-Recall Curve or ROC Curve generation might be used. The research of Minaee et al. (2021) suggested more popular metrics for evaluating the quality of the classification such as accuracy and error rate.

5. System Deployment

The cyberbullying detection model will be presented on a simple website. Users can input words, phrases, or sentences, and the model will promptly assess whether the content contains cyberbullying (binary classification, either cyberbullying or not-cyberbullying). If cyberbullying is detected, the model will also provide an analysis categorizing it into specific cyberbullying categories for a more detailed understanding of the content's nature (qualitative classification).

1.5.3 Constraint & Project Boundaries

Constraints:

1. The datasets used for this project is restricted to publicly available data due to privacy laws. For example, Kaggle.com.
2. The project will utilize existing open-source machine learning models or algorithms, eliminating the need for developing these components from scratch.
3. The detection models will be developed for English language content only.
4. The system will prioritize accuracy over real-time detection.

What will be done as part of the project:

1. The dataset used for this project will gather tweets on common social media platforms.
2. Cleaning and preprocessing the data to remove noise and irrelevant content.
3. Develop a detection model to identify cyberbullying patterns across major social media platforms within English language content.
4. The cyberbullying annotation system is designed to categorize cyberbullying content across various dimensions, including age, ethnicity, racism, gender, sexism, religion, aggression, and toxicity.
5. Conducting tests to assess accuracy, precision, recall, and F1-score.
6. Create a simple webpage for users to test use on the detection models.

What will not be done as part of the project:

1. The system will not support detection in languages other than English.
2. The project will not build a complete social media platform moderation tool. Its scope is confined to the detection of cyberbullying instances.
3. The cyberbullying detection system will aim to minimize false detections, but the complete elimination of errors is out of scope due to the nuances of human language and communication.
4. The system will not include real-time monitoring and intervention capabilities.

1.6 Potential Benefit

1.6.1 Tangible benefit

Tangible benefit refers to measurable outcomes or advantages that can be directly observed or quantified. The defining factors of this project include:

1. **Improved Safety**: By detecting and annotating instances of cyberbullying, platforms can take immediate action, such as removing harmful content or issuing warnings to users, thereby creating a safer online environment.
2. **Data Analytics**: Such systems can provide quantifiable data on the prevalence and patterns of cyberbullying, aiding in research and policymaking.
3. **Resource Efficiency**: Automated detection can reduce the need for manual moderation, saving time and labour costs.
4. **Compliance**: Some regions or platforms may have regulations or guidelines against cyberbullying. The detection system helps ensure compliance with these rules.
5. **User Retention**: By actively combating cyberbullying, platforms can increase user satisfaction and retention, as users are more likely to stay on platforms where they feel safe.

1.6.2 Intangible benefit

Intangible benefits refer to non-measurable outcomes or advantages that impact feelings, perceptions, or values, but cannot be easily quantified.

1. **Reputation Enhancement**: Platforms that actively combat cyberbullying are likely to be viewed more favourably by users and the public.
2. **Increased Trust**: Users are more likely to trust platforms that prioritize their safety and well-being.
3. **Emotional Well-being**: Reducing cyberbullying can lead to better mental and emotional health for users, as they experience less online harassment.
4. **Community Building**: A platform free from cyberbullying fosters a more positive and supportive community, encouraging constructive interactions.
5. **Promotion of Positive Behaviour**: By setting standards and norms against cyberbullying, platforms can influence users to engage in more positive online behaviours.

1.6.3 Target user

The cyberbullying detection and annotation system is suggested to use by these users:

1. **Individuals**: Individuals especially vulnerable groups such as children, elderly, or those with mental health issues, can reduce themselves from exposing to harmful content. They can engage in online activities without the fear of being bullied or harassed, leading to better mental and emotional well-being.

2. **Educational sector**: Schools and universities can ensure that their online platforms are free from cyberbullying, promoting a positive learning environment. The educational institutions can also do immediate action when detected cyberbullying.

3. **Social media platform**: Platforms that actively combat cyberbullying tend to increase user satisfaction and retention, more favourably by the public as they provide a peaceful community. Apply the cyberbullying detection system can also reduce the need for manual moderation, saving time and labour costs, benefit the platform itself.

1.7 Overview of the IR

The first chapter serves as the foundational segment of the report, outlining the project's essence and expected outcomes. It begins with an introduction, then pinpoints the problem and the project's aim. The chapter lists measurable objectives, defines the project's scope, and outlines its benefits. It also touches on the target users and concludes with the project plan.

Chapter 2, titled “Literature Review”, offers a thorough exploration of relevant literature to ground the project. It begins with an introduction, followed by domain research that covers broad topics to specific theories and technologies. The chapter then assesses similar systems, noting their features and efficacy. Technical research follows, detailing potential system requirements and technical tools. The chapter wraps up with a concise summary of its key insights.

Chapter 3, titled "Methodology," discusses the research and system development approach for the project. It commences with an introduction, followed by a detailed exploration of the chosen methodology. The chapter then delves into data collection, data understanding and data pre-processing. The chapter concludes with a summary, providing a cohesive understanding of the project's methodological framework.

The last chapter, “Conclusion”, wraps up the initial segment of the project. It begins by discussing the accomplishments of the project's first part. The chapter then evaluates the depth and adequacy of the research and investigation concerning the project's goals. Lastly, it addresses potential gaps in the research and design areas, suggesting areas for further exploration and enhancement. This chapter serves as a reflective checkpoint, assessing progress and identifying future directions.

1.8 Project Plan

Task Name	Duration	Start Date	Finish Date	Status
Project Proposal Form	10 days	Mon 11/9/23	Fri 22/9/23	Done
CHAPTER 1: INTRODUCTION	**5 days**	**Mon 2/10/23**	**Fri 6/10/23**	Done
1.1 Introduction	2 days	Mon 2/10/23	Tue 3/10/23	Done
1.2 Problem Background	2 days	Mon 2/10/23	Tue 3/10/23	Done
1.3 Project Aim	1 day	Mon 2/10/23	Mon 2/10/23	Done
1.4 Objectives	1 day	Mon 2/10/23	Mon 2/10/23	Done
1.5 Scope	2 days	Wed 4/10/23	Thu 5/10/23	Done
1.6 Potential Benefit	1 day	Fri 6/10/23	Fri 6/10/23	Done
1.6.1 Tangible Benefit	1 day	Fri 6/10/23	Fri 6/10/23	Done
1.6.2 Intangible Benefit	1 day	Fri 6/10/23	Fri 6/10/23	Done
1.6.3 Target User	1 day	Fri 6/10/23	Fri 6/10/23	Done
1.7 Overview of IR	1 day	Fri 6/10/23	Fri 6/10/23	Done
1.8 Project Plan	1 day	Fri 6/10/23	Fri 6/10/23	Done
CHAPTER 2: LITERATURE REVIEW	**14 days**	**Mon 9/10/23**	**Thu 26/10/23**	Done
2.1 Introduction	1 day	Mon 9/10/23	Mon 9/10/23	Done
2.2 Domain Research	4 days	Tue 10/10/23	Fri 13/10/23	Done
2.3 Similar Systems/Works	4 days	Mon 16/10/23	Thu 19/10/23	Done
2.4 Technical Research	4 days	Fri 20/10/23	Wed 25/10/23	Done
2.5 Summary	1 day	Thu 26/10/23	Thu 26/10/23	Done
CHAPTER 3: METHODOLOGY	**14 days**	**Fri 27/10/23**	**Wed 15/11/23**	Done
3.1 Introduction	1 day	Fri 27/10/23	Fri 27/10/23	Done
3.2 Methodology	4 days	Mon 30/10/23	Thu 2/11/23	Done
3.3 Data Collection and Data Understanding	4 days	Fri 3/11/23	Wed 8/11/23	Done
3.4 Initial Data Pre-processing	4 days	Thu 9/11/23	Tue 14/11/23	Done
3.5 Summary	1 day	Wed 15/11/23	Wed 15/11/23	Done
CHAPTER 4: CONCLUSION	**2 days**	**Thu 16/11/23**	**Fri 17/11/23**	Done
REFERENCES	1 day	Mon 20/11/23	Mon 20/11/23	Done
APPENDICES	1 day	Mon 20/11/23	Mon 20/11/23	Done

Figure 1: Project Plan of this project

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

In this chapter, we embark on a detailed exploration of the current landscape of cyberbullying detection, delving into domain research, an examination of existing systems, and a deep dive into the technical methodologies employed.

The domain research provides a comprehensive exploration into the domain of cyberbullying detection. It encompasses a detailed examination of text analysis techniques, their application in identifying and annotating instances of cyberbullying, and the pivotal role of Natural Language Processing (NLP) and feature extraction in processing textual data. The chapter also delves into various machine learning models that are instrumental in this field, highlighting their effectiveness and challenges. Additionally, it presents a comparative analysis of seven similar works, offering critical insights into the methodologies and findings of other researchers. On the technical side, the chapter discusses the rationale behind selecting Python for NLP tasks, the advantages of using Jupyter Notebook within Visual Studio Code as the development environment, and the pivotal roles of key Python libraries like Pandas and NLTK in data analysis and model development. This chapter thus lays the foundation for understanding both the theoretical framework and the practical tools and techniques essential for cyberbullying detection research.

Finally, the chapter concludes with a synopsis that encapsulates the key findings, linking back to the overarching themes of the domain's challenges, the insights gained from existing systems, and the technical underpinnings that support this vital work.

2.2 Domain Research

2.2.1 Text Analysis in Cyberbullying Detection

Text analysis is a machine learning method that enables the automated extraction of meaningful information from unstructured textual content. (MonkeyLearn, 2019) Text analysis could be an indispensable tool in the fight against cyberbullying, leveraging the vast amounts

of data generated in online communications to detect harmful behaviour. Text analysis encompasses a range of techniques from fundamental to complex, including text classification, information extraction, and word frequency analysis.

Text classification, also referred to as text categorization, is a fundamental challenge in the field of natural language processing. (Minaee et al., 2021) Text classification involves building models that categorize new documents into established categories. This process is intricate, encompassing more than just model training. It includes several preparatory steps such as data preprocessing, transformation, and dimensionality reduction. (Mirończuk & Protasiewicz, 2018) Text classification continues to be a significant area of study, employing a range of techniques and their integrations within multifaceted systems.

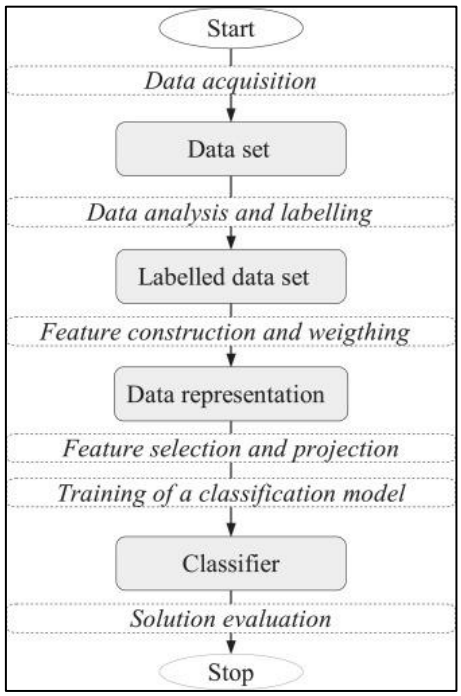


Figure 2: Flowchart of the text classification process. (Mirończuk & Protasiewicz, 2018)

Text classification strategies fall into two main groups: rule-based and machine learning-based approaches. (Minaee et al., 2021) Rule-based methods rely on predefined rules crafted from in-depth domain expertise. On the other hand, machine learning-based methods are better suited for identifying cyberbullying content because they categorize text by learning from data patterns.

2.2.2 Cyberbullying Detection and Annotation

An effective approach to creating a cyberbullying detection system involves the application of machine-based techniques, including machine learning, natural language processing (NLP) and deep learning. These methods are crucial for categorizing and analysing information effectively. (Bayari & Bensefia, 2021)

The research of Teng and Varathan (2023) presents the framework of constructing Binary Text Classification Cyberbullying Detection Model using two approaches – Conventional Machine Learning and Transfer Learning. The former approach implements feature extraction and machine learning model, and the latter approach utilizes pre-trained language models.

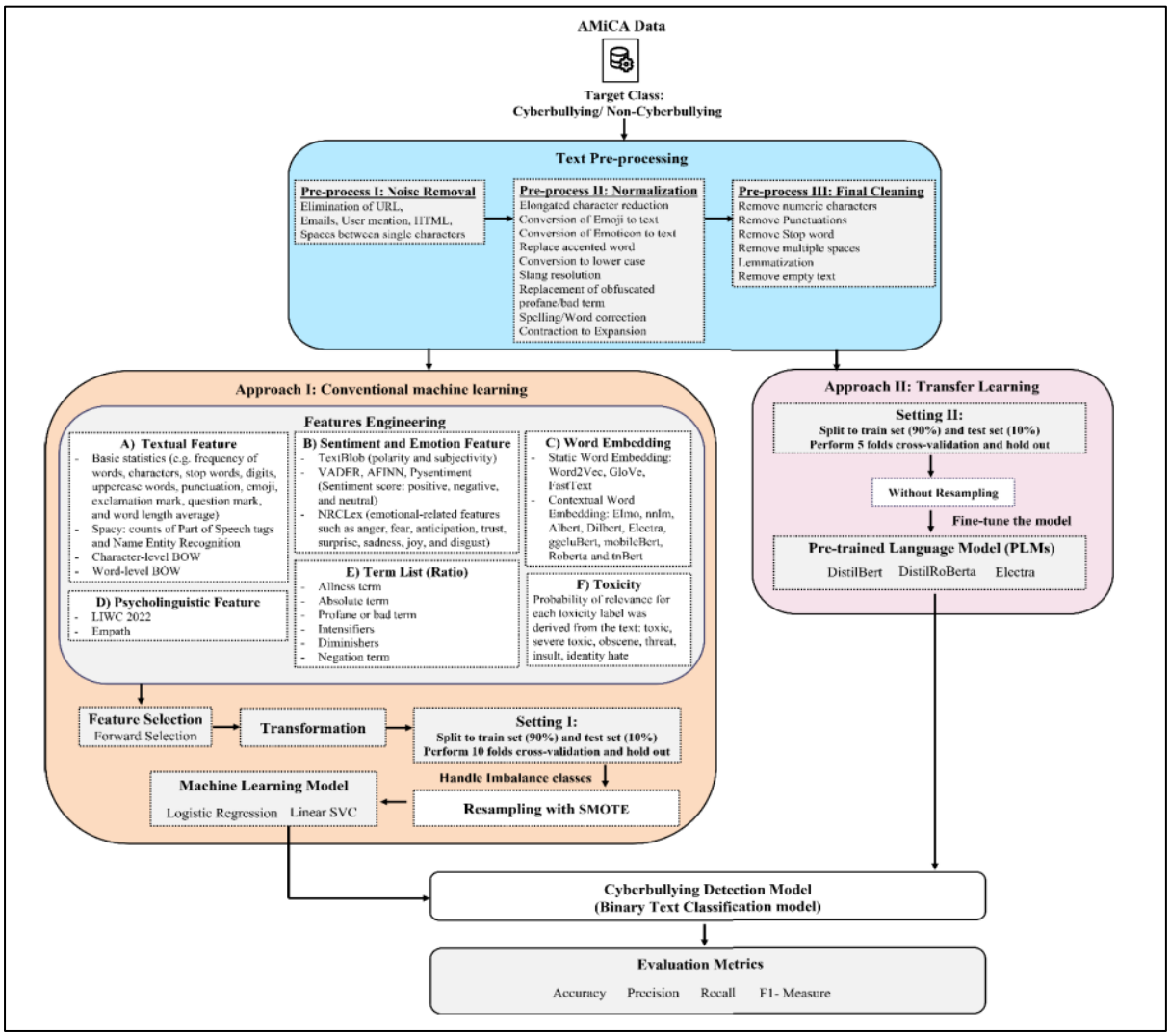


Figure 3: Overview of research framework on the model development of cyberbullying detection. (Teng & Varathan, 2023)

Annotating data within the context of social media presents significant challenges because of the rich and varied forms of communication that characterize these platforms. (Yi & Zubiaga, 2022) Social media conversations are multimodal, meaning they can include not just text, but images, videos, emojis, and audio. Each mode carries nuanced expressions of sentiment, sarcasm, humour, and other complex human emotions, which can be difficult to interpret consistently and accurately.

Cyberbullying, which includes online harassment and threats, has been linked to various factors such as age, gender, anonymity and so on. (Mkwananzi & Smuts, 2023) The significance of text analysis in this context lies in its capability to methodically detect and classify the language indicative of bullying.

Driver	Research Topic	Driver summary
Age	"Cyberbullying, University of Wisconsin Strout" (Aune, 2009)	Victims tend to suffer from cyberbullying in their adolescent and early adulthood years. Older students are likely to bully younger students.
	Hamal, 2017)	
	Poole, 2017)	
Gender	"The experience of victimization as the result of cyberbullying among college students" (Poole, 2017)	Girls are more likely to experience cyberbullying as victims; boys are more involved as bullies and victims.
	"Cyberbullying in schools: An examination of preservice teachers' perception" (Li, 2007)	
	Aune, 2009)	
	"Psychological, physical, and academic correlates of cyberbullying and traditional bullying" (Kowalski & Limber, 2013)	
	Hamal, 2017)	
	Poole, 2017)	
Family structure	"Cyberbullying from a socio-ecological perspective: A contemporary synthesis of findings from EU Kids Online" (Gorzig & Machackova, 2015)	The home environment of a child is a factor in cyberbullying. Family break-up is likely to cause involvement of a child in cyberbullying.
	Hamal, 2017)	
Race	"Cyberbullying in schools: An examination of preservice teachers' perception" (Li, 2007)	Children who bully usually come from families with low income and low education.
	Poole, 2017)	
Technology	"Cyber bullying an old problem in a new guise?" (Campbell, 2005)	Limited access to technology reduces cyberbullying and access to technology intensifies cyberbullying.
	"Cyberbullying in schools: An examination of preservice teachers' perception" (Li, 2007)	
	"Cyberbullying from a socio-ecological perspective: A contemporary synthesis of findings from EU Kids Online" (Gorzig & Machackova, 2015)	
Anonymity	"Cyberbullying: A Review of the Literature. Universal Journal of Educational Research" (Notar, Padgett, & Roden, 2013)	Anonymity creates instances where young people become wild and behave in a way they will never do when they are offline.
	Moore et al., 2012)	
	Poole, 2017)	
Traditional bullying	"Cyberbullying in schools: An examination of preservice teachers' perception" (Li, 2007)	Students involved in traditional bullying find it much easier to practise cyberbullying.
	Waasdorp & Bradshaw, 2015)	
Academic achievement	"Cyber bullying an old problem in a new guise?" (Campbell, 2005)	Cyber victims and bullies are likely to perform poorly at school
	"Cyberbullying in schools: An examination of preservice teachers' perception" (Li, 2007)	
	Kowalski & Limber, 2012)	
Awareness of cyber safety	"Cyber bullying an old problem in a new guise?" (Campbell, 2005)	Cyberbullying is a universal problem, but it varies from one country to another. Students from different countries and different cultures differ in terms of cyberbullying.
	"Cyberbullying in schools: An examination of preservice teachers' perception" (Li, 2007)	

Figure 4: Drivers of cyberbullying identified. (Mkwananzi & Smuts, 2023)

2.2.3 Natural Language Processing (NLP) and Feature Extraction

Natural language processing is a branch of artificial intelligence that deals with the interaction between computers and humans through natural language. It aims to parse, understand and derive meaning from human language in a structured way. (Tabassum & Patil, 2020) NLP can be applied in areas like text classification, information extraction, speech recognition, feature extraction etc. Before performing those tasks, a variety of techniques needed for handling the unstructured data. The paper of Tabassum & Patil (2020) highlighted the common pre-processing techniques, which include sentence segmentation, convert lowercase, tokenisation, POS Tagging, remove stop words, stemming and lemmatization.

To train the machine-learning based model, relevant training data, features and expected output have to be fed into the machine leaning algorithm to create the classification model for detecting the cyberbullying.

Feature extraction refers to the process of transforming raw text data into a set of numerical features that can be used by machine learning algorithms. (Wang et al., 2020) Feature extraction is indeed a technique that can be used for dimensionality reduction which reduces the number of random variables under consideration by obtaining a set of principal variables. It can be crucial for improving the efficiency of computational tasks, reducing the complexity of a model, and helping to alleviate the "curse of dimensionality" when dealing with high-dimensional data. (Pathare & Joshi, 2023)

Common textual features like Bag of Words (BoW) and word embedding are mentioned in the paper of Bayari & Bensefia (2021). These methods involve expanding a predefined list of offensive terms and assigning varying weights to these terms to uncover both overt and latent characteristics of bullying. The study of Putri Waisnawa et al. (2022) highlights the use of techniques like TF-IDF and N-grams for feature extraction in the pre-processing stage.

Bag of Words (BoW)

This method involves creating a vocabulary of all the unique words and then representing each document as a vector indicating the presence or frequency of each word. (Qader et al., 2019) BoW can result in high-dimensional feature spaces, particularly with large vocabularies, so

dimensionality reduction is often applied to focus on the most relevant words. The limitation of BoW is highlighted by Tabassum and Patil (2020) that it prioritizes words based on their frequency, assuming more frequent words are more important. However, some high-frequency words may not contribute significantly to classification or clustering tasks due to their low informational content. Additionally, longer documents tend to have inflated frequency counts, leading to potential inaccuracies in the BoW model. To address these drawbacks, the TFIDF (Term Frequency-Inverse Document Frequency) model is used as an alternative approach.

Word Embeddings

Word embedding models maps words and phrase into vectors of real numbers, effectively capturing both semantic and syntactic information. Many tasks can use the word similarity calculated by word embedding. (Li & Gong, 2021) Word embeddings can efficiently reduce feature space for text classification tasks. (Teng & Varathan, 2023) Techniques like Word2Vec create compact vector representations of words by mapping them into continuous vector spaces based on the context of their appearance. This reduces the dimensionality from potentially tens of thousands of binary BoW features to maybe a few hundred continuous features.

Term Frequency-Inverse Document Frequency (TF-IDF)

TF-IDF comprises two components: Term Frequency (TF) and Inverse Document Frequency (IDF). TF measures a word's frequency normalized by document length, while IDF reduces the weight of common words across documents, emphasizing unique terms. (Tabassum & Patil, 2020) TF-IDF is used to determine the significance of terms, resulting in a weight matrix as its output. The values in this matrix rise in accordance with the term frequency in TF-IDF but are moderated by the word's prevalence in the dataset. (Naeem et al., 2022)

2.2.4 Machine learning models

Most of the research commonly selected Naïve Bayes and Support Vector Machines (SVM) as the classification algorithm. Atoum (2021) utilizes Support Vector Machine (SVM), Naive Bayes classifiers and deep learning approaches to detect cyberbullying. The model demonstrated promising results, with SVM classifiers outperforming Naïve Bayes classifiers in effectiveness. Another research from Raj et al. (2021) proposes a novel neural network framework for cyberbullying detection, emphasizing the importance of feature extraction and

word-embedding techniques. The study compares various classification methods and feature extraction techniques, finding that bidirectional neural networks and attention models with specific feature extraction methods provide high classification results. The research highlights the significant role of feature extraction in enhancing the performance of cyberbullying detection models.

Naïve Bayes

Naïve Bayes model is a probabilistic algorithm based on Bayes' Theorem used for classification task. Calculate the probability of occurrence of the words of a sample text based on the words of a set of texts given. (Khurana et al., 2022)

Naïve Bayes model is selected as the baseline model for this project. As a commonly used algorithm, it serves as a standard or 'baseline' to compare against more complex models. If a more sophisticated model does not perform significantly better than the Naïve Bayes model, it might not be worth the additional complexity.

Support Vector Machines (SVM)

Support Vector Machines (SVMs) are a powerful and versatile class of supervised machine learning algorithms, primarily used for two-group classification. Word frequencies is the common feature used to classify texts using SVM. (Stecanella, 2017)

2.3 Similar Systems/Works

Table 1 presents an overview of existing algorithms and models for detecting cyberbullying in text, evaluating their effectiveness, strengths, and weaknesses, and includes case studies of successful implementations.

Research Topic	Author(s)	Description	Dataset Used	Analysis Techniques	Evaluation Techniques	Result / Outcomes	Limitations
Guidelines for Detecting Cyberbullying in Social Media Data Through Text Analysis	Mkwananzi and Smuts (2023)	The study aimed to pinpoint the factors contributing to cyberbullying and develop models to classify if social media text includes cyberbullying content.	-	Support Vector Machines (SVM) and Naïve Bayes models	Confusion matrix	The Naïve Bayes classifier was better than SVM in terms of classification accuracy.	This study focused solely on comparing SVM and Naïve Bayes classifiers. Hence, its findings are mainly applicable to situations involving these two classifiers and may not apply to other machine learning algorithms, requiring further research for broader generalization.
Cyberbullying Detection With Fairness Constraints	Gencoglu (2021)	Focus more on the measurement standard of cyberbullying detection for example gender, language, date or	Four publicly available datasets collected from four different media.	Deep neural network Multilingual language model, sentence-DistilBERT Adam optimizer	F1 Score Matthews correlation coefficient (MCC) McNemar's test (also known as within-	The integration of fairness constraints in neural network models for cyberbullying detection was found to be effective in reducing bias and improving detection	Setting different thresholds for each group might lead to more equitable models but also increases the complexity of model tuning.

		recency and hate speech on the basis of religion, race, and nationality.			subjects chi-squared test) Sum of FNED and FPED	performance across various platforms and groups.	
Cyberbullying Detection on Twitter Using Deep Learning-Based Attention Mechanisms and Continuous Bag of Words Feature Extraction	Fati et al. (2023)	This study focuses on detecting cyberbullying on Twitter using deep learning-based attention mechanisms and continuous bag of words (CBOW) feature extraction.	The research utilized a dataset comprising tweets collected from Twitter. These tweets were preprocessed and labeled for cyberbullying content.	Deep learning models, attention mechanisms Continuous Bag of Words (CBOW) for feature selection	Evaluated using standard metrics like accuracy, precision, recall, and F1-score. The performance of the attention-based models was compared with traditional machine learning models and other deep learning approaches.	The research found that the attention-based deep learning models, combined with CBOW feature extraction, significantly improved the detection of cyberbullying on Twitter. These models outperformed traditional machine learning methods and other deep learning approaches in terms of accuracy and other metrics.	The study acknowledges limitations, such as dependency on the quality and size of the dataset, possible errors in data or generalizability for a model to other social media platforms or text types. It has also pointed out that understanding deep learning models requires further research, which will be needed in order to improve model transparency and understandability.
Improving cyberbullying detection using Twitter users' psychological features and	Balakrishnan et al. (2020)		Twitter dataset that includes 5453 tweets collected with the hashtag #Gamergate	Naïve Bayes, Random Forest and J48 were used to classify the tweets into one of four categories: bully, aggressor, spammer and normal.	Performances metrics includes F-measure, accuracy and ROC curve. Confusion matrix Root mean squared error	The results indicated that the detection of cyberbullying improved when incorporating personalities and sentiments. The integrated model, which combined the key features, produced the best detection accuracy.	The study primarily focuses on Twitter data, and the results might not be generalizable to other social media platforms. Additionally, while the study integrates psychological features for detection, the complexity of human behaviour

machine learning					T-tests		and the evolving nature of online interactions might present challenges in real-world applications.
Cyberbullying Detection Neural Networks using Sentiment Analysis	Atoum (2020)	The study proposes a Sentiment Analysis (SA) model for identifying cyberbullying texts on Twitter using machine learning classification tools.	Twitter data	Support Vector Machines (SVM) and Naïve Bayes (NB) classifiers		The model showed encouraging outcomes, with SVM classifiers performing better than NB classifiers.	The study does not specify the size and diversity of the dataset, which could affect the generalizability of the results.
Approaches to Automated Detection of Cyberbullying : A Survey	Salawu et al. (2020)	This paper presents a systematic review of automated approaches to cyberbullying detection, categorizing them into supervised learning, lexicon-based, rule-based, and mixed-initiative approaches.	Survey	Supervised learning (e.g., SVM, NB), lexicon-based, rule-based, mixed-initiative approaches	Literature review and categorization	Identification of key challenges such as lack of labelled datasets and non-holistic consideration of cyberbullying.	As a survey, it does not provide empirical results from a specific dataset or analysis technique.

Polarity and Intensity: the Two Aspects of Sentiment Analysis	Tian et al. (2018)	The research decomposes sentiment scores into polarity and intensity and studies how these are conveyed through individual modalities and combined multimodal models.		Unimodal and multimodal multi-task learning models	Multi-task learning with sentiment score prediction as the main task and polarity/intensity classification as auxiliary tasks	Sentiment analysis benefits from multi-task learning, with modalities differing in conveying polarity and intensity.	The abstract does not detail the specific context of cyberbullying or the types of datasets used, which are crucial for applicability in cyberbullying detection.

Table 1: Similar works of cyberbullying detection system.

2.4 Technical Research

2.4.1 Programming language chosen – Python

Python has become the most popular choice for machine learning and natural language processing (NLP). The study of Sultonov (2023) emphasises the importance of Python in the field of machine learning, highlighting the simple and straightforward syntax of Python, which allows programmers to focus on logic and algorithms rather than complicated syntax.

A major factor in Python's suitability for cyberbullying detection is its extensive selection of libraries that offer supporting functions for both machine learning and natural language processing tasks. (CFI Team, n.d.) Flexible machine learning models is easy to build on a minimum programming effort have been provided through libraries like TensorFlow and PyTorch which can be used for creating detailed models, able to understand human language's subtleties. PyTorch (n.d.) share models through a repository dedicated to pre-trained models, while TensorFlow (n.d.) publish models to a pre-trained model repository designed for research exploration.

For natural language processing, libraries like NLTK and spaCy offer tools for text processing and linguistic analysis that are essential for interpreting and classifying textual data, which is a significant component of cyberbullying detection. These libraries include also pre-trained models on large corpora of text, which can be fine-tuned on domain-specific data to improve the accuracy of cyberbullying detection. (NLTK, 2023)

Moreover, Python’s Scikit-learn library is widely used for implementing traditional machine learning algorithms which can be effectively employed in feature engineering and classification tasks within a cyberbullying context. Scikit-learn offers functionality for dividing datasets into training and testing sets, implementing various feature search and preprocessing algorithms, and constructing models based on these algorithm parameters. It also provides metrics like accuracy for evaluating model performance, allows for parameter adjustments, and ultimately facilitates the creation of a trained model (Sultonov, 2023).

Additionally, Python's data handling libraries such as Pandas and NumPy are indispensable for preprocessing and manipulating large datasets that are characteristic of online interactions. (Codemotion, 2023) This interoperability between data manipulation and machine learning libraries streamlines the pipeline from raw data collection to model deployment in cyberbullying detection systems.

Word cloud visualization in Python is a compelling way to visually represent text data. It involves creating a 'cloud' of words, where the size of each word indicates its frequency or importance in the dataset. Python offers libraries like wordcloud and matplotlib to generate these visualizations. The generated word cloud can then be displayed using matplotlib, providing an intuitive and visually appealing representation of text data, often used to highlight the most frequent or significant words in a text corpus.

By leveraging these libraries, researchers can efficiently experiment with different machine learning paradigms, from supervised learning (labelled data) to unsupervised learning (unlabelled data), which is critical in identifying emerging forms of cyberbullying.

2.4.2 IDE chosen – Jupyter Notebook with Visual Studio Code

Working with Jupyter Notebook in Visual Studio Code is powerful for researchers working on project especially when collaborating is not necessary. Jupyter Notebook supports Markdown cells and Code cell, providing programmers interactivity feedback by execute the code in small block with immediate visualization. It allows programmers to style the text in Markdown cells, including headers, bulleted lists and font style. (Driscoll, 2018)

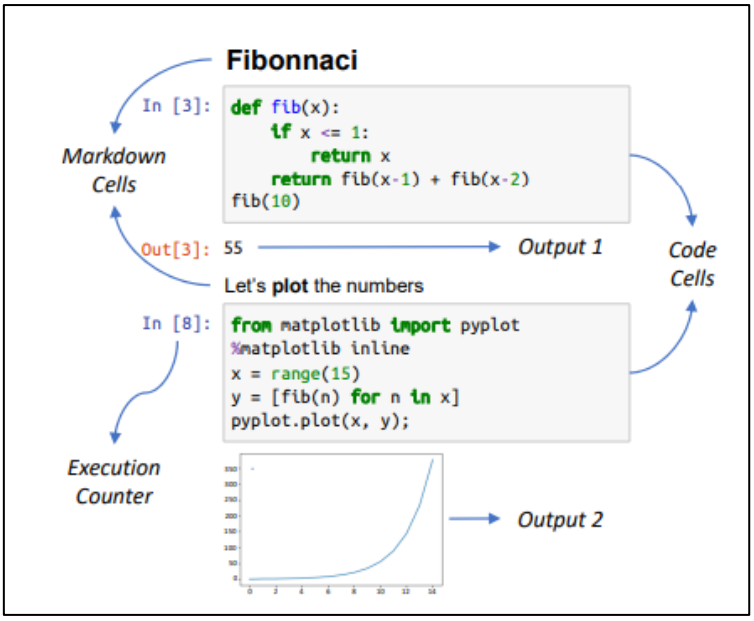


Figure 5: A notebook example with markdown, code and output (Pimentel et al., 2019).

By utilizing Visual Studio Code, programmers benefit from a sophisticated code editing experience coupled with the interactivity of Jupyter Notebooks, all within a single, integrated environment. This setup is ideal for who more comfortable with Visual Studio Code's extensive features, such as IntelliSense and version control. IntelliSense offers functionalities such as auto-completions, member lists, quick information on methods, and hints for parameters. This enables programmers to maintain high productivity levels, matching the efficiency they experience in the code editor. (Visual Studio Code, 2021) Moreover, running Jupyter locally through Visual Studio Code provides greater control over computational resources, enabling efficient handling of demanding tasks without the constraints of cloud-based platforms like Google Colab. This choice ensures that the development process is tailored to individual workflow preferences.

2.5 Summary

The literature review provides an in-depth analysis of cyberbullying detection, covering domain research, existing systems, and technical methodologies. The domain research section delves into text analysis techniques, highlighting their role in cyberbullying detection through Natural Language Processing (NLP) and feature extraction. It examines various machine learning models, discussing their effectiveness and challenges, and compares seven similar works to provide insights into different methodologies.

The chapter also explores the technical aspects of cyberbullying detection, emphasizing the selection of Python as the programming language due to its simplicity and the availability of libraries like TensorFlow, PyTorch, NLTK, and spaCy. These libraries facilitate the creation of models capable of understanding the subtleties of human language and are crucial for text processing and linguistic analysis. The chapter discusses the use of Jupyter Notebook within Visual Studio Code as the development environment, highlighting its benefits for code editing and interactivity.

Key topics include text classification, cyberbullying detection and annotation, NLP and feature extraction, and machine learning models. The text analysis section discusses the importance of text classification in NLP, covering rule-based and machine learning-based methods. The cyberbullying detection and annotation section focuses on machine learning, NLP, and deep learning methods for categorizing and analyzing information. The NLP and feature extraction section details the process of transforming raw text data into numerical features for machine learning algorithms, discussing techniques like Bag of Words, word embeddings, and TF-IDF. The machine learning models section reviews the use of Naïve Bayes and Support Vector Machines in cyberbullying detection.

Finally, the chapter presents a comparative analysis of similar systems and works, evaluating their effectiveness, strengths, and weaknesses in detecting cyberbullying. This comprehensive review lays a solid foundation for understanding the theoretical and practical aspects of cyberbullying detection research.

CHAPTER 3: METHODOLOGY

3.1 Introduction

This chapter introduces the methodology used for this data analysis project, which is CRISP-DM. A clear methodology should be determined to make sure that the project is conducted in a proper flow. CRISP-DM provides a clear procedure for data analysis project.

Then, the project progresses into stages of data collection, comprehension, and preprocessing, aligning with the established methodological guidelines. The data understanding phase involves loading and exploring the gathered datasets, followed by a selective process to identify and retain only the pertinent data for further processing. The data preprocessing phase encompasses a series of steps aimed at refining the text data, preparing it for the feature extraction process which is slated for execution in Part 2 of this project.

Finally, conclude the processes undertaken in this phase of the project. The refined dataset is a critical output of this phase, representing a clean, organized, and meaningful compilation of data, specifically used to proceed the subsequent tasks in Part 2 of the project.

3.2 Methodology

3.2.1 Introduction of CRISP-DM Methodology

The Cross Industry Standard Process for Data Mining (CRISP-DM) is the common framework for data mining project planning. CRISP-DM provides a structured approach to planning, organizing and implementing data mining projects across various industries. The methodology is cyclical in nature, ensuring continuous improvement in data analysis through iterative processes. (Schröer et al., 2021) CRISP-DM consists of six iterative processes: Business Understanding, Data Understanding, Data Preparation, Modelling, Evaluation and Deployment.

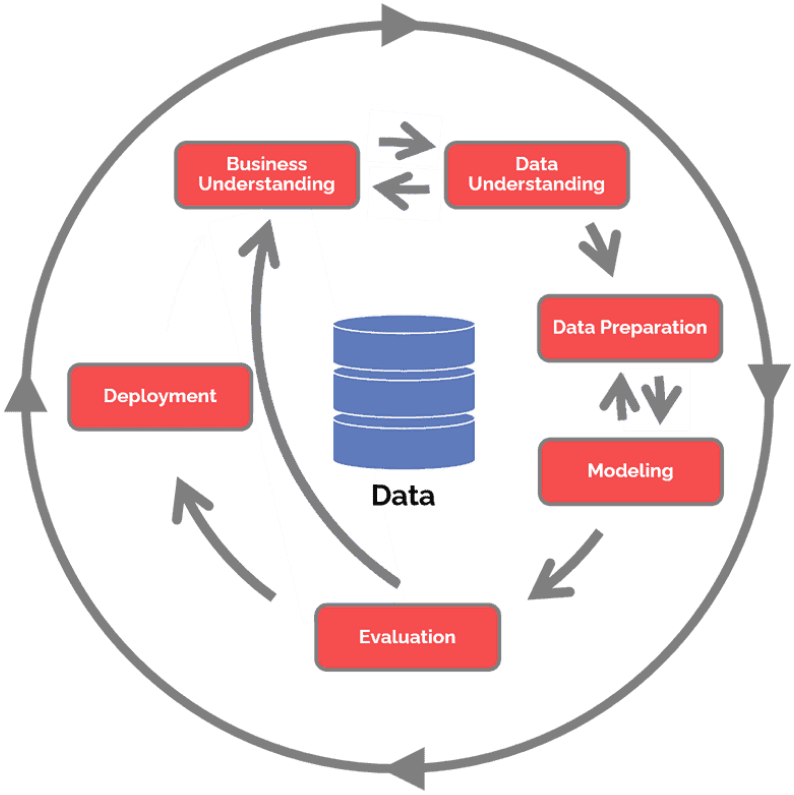


Figure 6: CRISP-DM Diagram (Hotz, 2023)

3.2.2 Methodology choice and justification

The selected methodology for this project is CRISP-DM is because of its structured approach. CRISP-DM provides a clear, step-by-step process that guides users from understanding the business problem to deploying the solution, ensuring a systematic approach to data mining. Although this methodology follows on step-by-step process, the cyclical design of CRISP-DM emphasizes iterative refinement, allowing for continuous improvement based on results and feedback. This ensures that the project remains aligned with business objectives and can adapt to new insights or changes.

Compared to other methodology, SEMMA is not suitable for this project as it does not have development stage. It also limited to utilize the SAS tools for best results. Additionally, KDD, known as Knowledge Discovery in Databases, emphasized on technical aspects of knowledge discovery, less focus on business context compared to CRISP-DM.

3.2.3 Phases of CRISP-DM

Process 1: Business Understanding

The first phase of CRISP-DM is to understand the project's objectives and requirements. The goal of this stage is to highlight important factors that could affect the deliverables of the project. (Smart Vision Europe, 2020) The scope and potential constraints should be clearly stated out in the project.

Process 2: Data Understanding

The focus in this stage is data gathering, explore the data collected and make sure the quality of data is readily available to use in this project. (Hotz, 2023) Necessary data processing steps might be conducted in this stage to understand the data.

Process 3: Data Preparation

This stage involves all tasks required to construct the final dataset, which might include data selection, data cleaning and data transformation. The selected dataset will be identified and the reason of selection will be recorded. (Hotz, 2023) The data cleaning is the most complex task when preparing the data. Finally, the datasets will be combined and format the data if needed.

Process 4: Modelling

Training, validating, and testing multiple models and algorithms are beneficial for contrasting the outcomes of various models during assessment. (Schröer et al., 2021) To build the models, the modelling techniques need to be determined at first. Then, split the datasets into training, validation and testing sets based on the prepared data. After training the models, interpret the model results and tuning the parameters for optimal performance.

Process 5: Evaluation

In this phase, the models will be evaluated and compared how well the models align with business objectives. Those that fulfil the chosen criteria are recognized as the approved models. (Smart Vision Europe, 2020) Then, review the models by conducting a more comprehensive examination of the data mining process to identify any crucial elements or tasks that might have been missed. Based on previous tasks, decide whether to move forward with deployment, continue refining, or start new projects.

Process 6: Deployment

Depending on the requirements, the development phase can involve generating a final report or integrating the model into business applications. Hence, plan the deployment process before commencing of deployment including the monitoring and maintenance plan. Lastly, the final output should be clearly documented. (Hotz, 2023)

3.3 Data Collection

The data collected is all from Kaggle.com but different datasets. The Cyberbullying Dataset includes nine datasets. All datasets are basically in csv file.

Dataset Name	Source (from Kaggle.com)
cyberbullying_tweets	Cyberbullying Classification
aggression_parsed_dataset	Cyberbullying Dataset
attack_parsed_dataset	Cyberbullying Dataset
kaggle_parsed_dataset	Cyberbullying Dataset
toxicity_parsed_dataset	Cyberbullying Dataset
twitter_parsed_dataset	Cyberbullying Dataset
twitter_racism_parsed_dataset	Cyberbullying Dataset
twitter_sexism_parsed_dataset	Cyberbullying Dataset
youtube_parsed_dataset	Cyberbullying Dataset

Table 2: Kaggle Sources of the datasets.

3.4 Data Understanding

This project requires two datasets: one for cyberbullying detection to discern if tweets contain cyberbullying elements, and another for categorizing the type of cyberbullying by attributes such as age, gender, or religion.

The preliminary step in handling these datasets involves comprehensively analysing each one to ascertain their relevance and applicability to the project's objectives.

3.4.1 Load and Explore Datasets

Import and explore the datasets by using Pandas Python Library. The figures below show the source code and output of loading and understanding one of the datasets. This understanding will be performed on each dataset. The results are showed in Table 3 for better understanding.

```
# import dataset
filepath = "C:/Users/user/OneDrive/year3-sem1/FYP/DATASET/cyberbullying_tweets.csv"
data = pd.read_csv(filepath)
```

Figure 7: Source code of loading dataset.

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 47692 entries, 0 to 47691
Data columns (total 2 columns):
 #   Column              Non-Null Count  Dtype 
---  ------              --------------  ----- 
 0   tweet_text          47692 non-null  object
 1   cyberbullying_type  47692 non-null  object
dtypes: object(2)
memory usage: 745.3+ KB
```

Figure 8: Source code and result of getting information about the dataset.

```
data.cyberbullying_type.value_counts()
```

```
cyberbullying_type
religion               7998
age                    7992
gender                 7973
ethnicity              7961
not_cyberbullying      7945
other_cyberbullying    7823
Name: count, dtype: int64
```

Figure 9: Source code and result of getting the number of tweets for each cyberbullying type.

Dataset Name	Dataset Size	The Proportion of Cyberbullying	Labels / Annotation	Observations
cyberbullying_tweets	47,692	Cyberbullying: 39,747 Not-cyberbullying: 7,945 cyberbullying_type religion 7998 age 7992 gender 7973 ethnicity 7961 not_cyberbullying 7945 other_cyberbullying 7823	Almost 8,000 tweets for each cyberbullying category (religion, age, gender, ethnicity)	Contain 2 columns, tweet text and type of cyberbullying. <class 'pandas.core.frame.DataFrame'> RangeIndex: 47692 entries, 0 to 47691 Data columns (total 2 columns): # Column Non-Null Count Dtype --- ------ -------------- ----- 0 tweet_text 47692 non-null object 1 cyberbullying_type 47692 non-null object dtypes: object(2) memory usage: 745.3+ KB
aggression_parsed_dataset	115,864	Cyberbullying: 14,782 Not-cyberbullying: 101,082 oh_label 0 101082 1 14782	-	The columns ed_label_0 and ed_label_1 might be the intermediate result when generating the dataset. Both columns will not be used if the records are selected. <class 'pandas.core.frame.DataFrame'> RangeIndex: 115864 entries, 0 to 115863 Data columns (total 5 columns): # Column Non-Null Count Dtype --- ------ -------------- ----- 0 index 115864 non-null int64 1 Text 115864 non-null object 2 ed_label_0 115864 non-null float64 3 ed_label_1 115864 non-null float64 4 oh_label 115864 non-null int64 dtypes: float64(2), int64(2), object(1) memory usage: 4.4+ MB

attack_parsed_dataset	115,864	Cyberbullying: 13,590
		Not-cyberbullying: 102,274
		`oh_label`
		`0 102274`
		`1 13590`
			-	The observation is same as the dataset above.
				`<class 'pandas.core.frame.DataFrame'>`
				`RangeIndex: 115864 entries, 0 to 115863`
				`Data columns (total 5 columns):`
				` # Column Non-Null Count Dtype`
				`--- ------ -------------- -----`
				` 0 index 115864 non-null int64`
				` 1 Text 115864 non-null object`
				` 2 ed_label_0 115864 non-null float64`
				` 3 ed_label_1 115864 non-null float64`
				` 4 oh_label 115864 non-null int64`
				`dtypes: float64(2), int64(2), object(1)`
				`memory usage: 4.4+ MB`

kaggle_parsed_dataset	8,799	Cyberbullying: 5,993
		Not-cyberbullying: 2,806
		`oh_label`
		`0 5993`
		`1 2806`
			-	Consist of two useful columns for this project are text and label.
				`<class 'pandas.core.frame.DataFrame'>`
				`RangeIndex: 8799 entries, 0 to 8798`
				`Data columns (total 4 columns):`
				` # Column Non-Null Count Dtype`
				`--- ------ -------------- -----`
				` 0 index 8799 non-null int64`
				` 1 oh_label 8799 non-null int64`
				` 2 Date 7557 non-null object`
				` 3 Text 8799 non-null object`
				`dtypes: int64(2), object(2)`
				`memory usage: 275.1+ KB`

<table>
<tr>
<td>toxicity_parsed_dataset</td>
<td>159,686</td>
<td>Cyberbullying: 15,362
Not-cyberbullying: 144,324

oh_label	
0	144324
1	15362

</td>
<td>-</td>
<td>Similar observation as attack_parsed_dataset

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 159686 entries, 0 to 159685
Data columns (total 5 columns):
 #   Column      Non-Null Count   Dtype
---  ------      --------------   -----
 0   index       159686 non-null  int64
 1   Text        159686 non-null  object
 2   ed_label_0  159686 non-null  float64
 3   ed_label_1  159686 non-null  float64
 4   oh_label    159686 non-null  int64
dtypes: float64(2), int64(2), object(1)
memory usage: 6.1+ MB
```

.</td>
</tr>
<tr>
<td>twitter_parsed_dataset</td>
<td>16,851</td>
<td>Cyberbullying: 5,347
Not-cyberbullying: 11,501

oh_label	
0.0	11501
1.0	5347

</td>
<td>The tweets that detected contain cyberbullying are further classify into sexism and racism.

Annotation	
none	11501
sexism	3377
racism	1970

</td>
<td>This dataset includes a column specify the cyberbullying class.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 16851 entries, 0 to 16850
Data columns (total 5 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   index       16851 non-null  object
 1   id          16850 non-null  object
 2   Text        16850 non-null  object
 3   Annotation  16848 non-null  object
 4   oh_label    16848 non-null  float64
dtypes: float64(1), object(4)
memory usage: 658.4+ KB
```

</td>
</tr>
</table>

twitter_racism_parsed_dataset	13,471	Cyberbullying: 1,970
		Not-cyberbullying: 11,501
		`oh_label`
		`0 11501`
		`1 1970`
			Contains only racism cyberbullying comments.
			`Annotation`
			`none 11501`
			`racism 1970`
				The observation is same as the dataset above.
				`<class 'pandas.core.frame.DataFrame'>`
				`RangeIndex: 13471 entries, 0 to 13470`
				`Data columns (total 5 columns):`
				` # Column Non-Null Count Dtype`
				`--- ------ -------------- -----`
				` 0 index 13471 non-null float64`
				` 1 id 13471 non-null float64`
				` 2 Text 13471 non-null object`
				` 3 Annotation 13471 non-null object`
				` 4 oh_label 13471 non-null int64`
				`dtypes: float64(2), int64(1), object(2)`
				`memory usage: 526.3+ KB`

twitter_sexism_parsed_dataset	14,881	Cyberbullying: 3,377
		Not-cyberbullying: 11,501
		`oh_label`
		`0.0 11501`
		`1.0 3377`
			Contains only sexism cyberbullying comments.
			`Annotation`
			`none 11501`
			`sexism 3377`
				The observation is same as the dataset above.
				`<class 'pandas.core.frame.DataFrame'>`
				`RangeIndex: 14881 entries, 0 to 14880`
				`Data columns (total 5 columns):`
				` # Column Non-Null Count Dtype`
				`--- ------ -------------- -----`
				` 0 index 14881 non-null object`
				` 1 id 14880 non-null object`
				` 2 Text 14880 non-null object`
				` 3 Annotation 14878 non-null object`
				` 4 oh_label 14878 non-null float64`
				`dtypes: float64(1), object(4)`
				`memory usage: 581.4+ KB`

youtube_parsed_dataset	3464	Cyberbullying: 3,377
		Not-cyberbullying: 11,501
		`oh_label`
		`0 3047`
		`1 417`
			-	This dataset contains columns that provide further information on each tweets data.
				`<class 'pandas.core.frame.DataFrame'>`
				`RangeIndex: 3464 entries, 0 to 3463`
				`Data columns (total 10 columns):`
				`# Column Non-Null Count Dtype`
				`0 index 3464 non-null int64`
				`1 UserIndex 3464 non-null object`
				`2 Text 3464 non-null object`
				`3 Number of Comments 3464 non-null int64`
				`4 Number of Subscribers 3464 non-null int64`
				`5 Membership Duration 3464 non-null int64`
				`6 Number of Uploads 3464 non-null int64`
				`7 Profanity in UserID 3464 non-null int64`
				`8 Age 3464 non-null int64`
				`9 oh_label 3464 non-null int64`
				`dtypes: int64(8), object(2)`
				`memory usage: 270.8+ KB`

Table 3: Outputs and observation of initial data understanding on datasets.

3.4.2 Comparing Datasets

Then, perform further understanding as there are some datasets look like having similar characteristics and results. Using the function of datacompy python library, compare the datasets. (Darshanaawalgaonkar, 2021) The datasets performed compare are shown in Table 4.

Comparison	Dataset 1	Dataset 2
1	twitter_parsed	twitter_racism_parsed
2	twitter_parsed	twitter_ sexism_parsed
3	aggression_parsed	attack_parsed

Table 4: Datasets performed comparison.

Comparison 1:

The number of rows with all compared columns equal is 13,471 which is same as the total number of rows of the twitter_racism_parsed_dataset. It can be inferred that the dataset is a subset of twitter_parsed_dataset.

```
# compare twitter_parsed and twitter_racism_parsed datasets
compare = datacompy.Compare(
    data5,
    data6,
    join_columns="Text",
    df1_name="twitter_parsed",
    df2_name="twitter_racism_parsed",
)
print(compare.report())
```

Figure 10: Source code of comparing twitter_parsed and twitter_racism_parsed datasets.

```
Row Summary
-----------

Matched on: text
Any duplicates on match values: No
Absolute Tolerance: 0
Relative Tolerance: 0
Number of rows in common: 13,471
Number of rows in twitter_parsed but not in twitter_racism_parsed: 3,380
Number of rows in twitter_racism_parsed but not in twitter_parsed: 0

Number of rows with some compared columns unequal: 0
Number of rows with all compared columns equal: 13,471
```

Figure 11: Report of comparing twitter_parsed and twitter_racism_parsed datasets.

Comparison 2:

Same as the comparison above, the results show that both the twitter_racism_parsed_dataset and twitter_sexism_parsed_dataset is derived from twitter_parsed_dataset. Hence, both datasets (twitter_racism_parsed and twitter_sexism_parsed) will not be used.

```
# compare twitter_parsed and twitter_sexism_parsed datasets
compare = datacompy.Compare(
    data5,
    data7,
    join_columns="Text",
    df1_name="twitter_parsed",
    df2_name="twitter_sexism_parsed",
)
print(compare.report())
```

Figure 12: Source code of comparing twitter_parsed and twitter_ sexism_parsed datasets.

```
Row Summary
-----------

Matched on: text
Any duplicates on match values: No
Absolute Tolerance: 0
Relative Tolerance: 0
Number of rows in common: 14,881
Number of rows in twitter_parsed but not in twitter_sexism_parsed: 1,970
Number of rows in twitter_sexism_parsed but not in twitter_parsed: 0

Number of rows with some compared columns unequal: 0
Number of rows with all compared columns equal: 14,881
```

Figure 13: Report of comparing twitter_parsed and twitter_ sexism_parsed datasets.

Comparison 3:

As long as the dataset size and the proportion of cyberbullying of the aggression_parsed_dataset and attack_parsed_dataset have high similarities, a necessary of comparing both datasets also needed.

```
# compare aggression_parsed and attack_parsed datasets
compare = datacompy.Compare(
    data1,
    data2,
    join_columns="Text",
    df1_name="aggression_parsed",
    df2_name="attack_parsed",
)
print(compare.report())
```

Figure 14: Source code of comparing aggression_parsed and attack_parsed datasets.

```
Row Summary
-----------

Matched on: text
Any duplicates on match values: Yes
Absolute Tolerance: 0
Relative Tolerance: 0
Number of rows in common: 115,864
Number of rows in aggression_parsed but not in attack_parsed: 0
Number of rows in attack_parsed but not in aggression_parsed: 0

Number of rows with some compared columns unequal: 33,495
Number of rows with all compared columns equal: 82,369

Column Comparison
-----------------

Number of columns compared with some values unequal: 3
Number of columns compared with all values equal: 2
Total number of values which compare unequal: 68,894
```

Figure 15: Report of comparing aggression_parsed and attack_parsed datasets.

```
data1.tail()
```
✓ 0.0s

	index	text	ed_label_0	ed_label_1	oh_label
115859	115859	`These sources don't exactly exude a sense ...	0.777778	0.222222	0
115860	115860	The Institute for Historical Review is a pee...	1.000000	0.000000	0
115861	115861	:The way you're trying to describe it in this...	1.000000	0.000000	0
115862	115862	== Warning == There is clearly a protection...	0.500000	0.500000	0
115863	115863	Alternate option=== Is there perhaps enough ne...	1.000000	0.000000	0

```
data2.tail()
```
✓ 0.0s

	index	text	ed_label_0	ed_label_1	oh_label
115859	115859	`These sources don't exactly exude a sense ...	0.888889	0.111111	0
115860	115860	The Institute for Historical Review is a pee...	0.900000	0.100000	0
115861	115861	:The way you're trying to describe it in this...	1.000000	0.000000	0
115862	115862	== Warning == There is clearly a protection...	0.800000	0.200000	0
115863	115863	Alternate option=== Is there perhaps enough ne...	1.000000	0.000000	0

Figure 16: Sample data of aggression_parsed and attack_parsed datasets.

3.4.3 Data Gathering

Based on the data understanding, the steps to be processed are determined to generate an initial dataset.

Dataset Name	Selecting data and pre-processing steps determined	Annotation
cyberbullying_tweets	- Use all two columns originally in dataset. - Add a new column storing binary representation (1 and 0) denoting cyberbullying or non-cyberbullying by transforming "cyberbullying_type" column.	age, ethnicity, gender, religion, other
aggression_parsed_dataset attack_parsed_dataset	- Join the dataset and remove the duplicate. - Add a new "aggression" column, insert data with "aggression". - Check is there any duplicates in both dataset.	aggression
kaggle_parsed_dataset	- Select "Text" and "oh_label" columns.	-
toxicity_parsed_dataset	- Select "Text" and "oh_label" columns. - Add a new "toxicity" column, insert data with "aggression".	toxicity
twitter_parsed_dataset	- Select "Text", "Annotation" and "oh_label" columns.	sexism, racism
twitter_racism_parsed_dataset	-	-
twitter_sexism_parsed_dataset	-	-
youtube_parsed_dataset	- Select "Text" and "oh_label" columns.	-

Table 5: Decision of selecting data.

a. Selecting columns

As mentioned in Table 5, select only the useful columns in the datasets. Another method that produces same result as above is dropping the unused column.

```
data1 = data1[["text", "oh_label"]]
data2 = data2[["text", "oh_label"]]
data3 = data3[["Text", "oh_label"]]
data4 = data4[["Text", "oh_label"]]
data5 = data5[["text", "oh_label", "annotation"]]
data8 = data8[["Text", "oh_label"]]
```

Figure 17: Source code of selecting columns from datasets.

```
Dataset  Index(['tweet_text', 'cyberbullying_type'], dtype='object')
Dataset1  Index(['text', 'oh_label'], dtype='object')
Dataset2  Index(['text', 'oh_label'], dtype='object')
Dataset3  Index(['Text', 'oh_label'], dtype='object')
Dataset4  Index(['Text', 'oh_label'], dtype='object')
Dataset5  Index(['text', 'oh_label', 'annotation'], dtype='object')
Dataset8  Index(['Text', 'oh_label'], dtype='object')
```

Figure 18: Summary of selecting columns from datasets.

b. Rename columns

Renaming the selected columns for consistency.

```
data.rename(
    columns={"tweet_text": "text", "cyberbullying_type": "cb_type"}, inplace=True
)
data1.rename(columns={"oh_label": "cb_label"}, inplace=True)
data2.rename(columns={"oh_label": "cb_label"}, inplace=True)
data3.rename(columns={"Text": "text", "oh_label": "cb_label"}, inplace=True)
data4.rename(columns={"Text": "text", "oh_label": "cb_label"}, inplace=True)
data5.rename(columns={"oh_label": "cb_label", "annotation": "cb_type"}, inplace=True)
data8.rename(columns={"Text": "text", "oh_label": "cb_label"}, inplace=True)
```

Figure 19: Source code of renaming columns.

```
Dataset  Index(['text', 'cb_type'], dtype='object')
Dataset1  Index(['text', 'cb_label'], dtype='object')
Dataset2  Index(['text', 'cb_label'], dtype='object')
Dataset3  Index(['text', 'cb_label'], dtype='object')
Dataset4  Index(['text', 'cb_label'], dtype='object')
Dataset5  Index(['text', 'cb_label', 'cb_type'], dtype='object')
Dataset8  Index(['text', 'cb_label'], dtype='object')
```

Figure 20: Source code of renaming columns.

c. Add new column

To ensure uniformity across all datasets, it's necessary to standardize their structure by adding new columns and populating them with appropriate data. This process involves adding a new column, "cb_label," to the "cyberbullying_tweets" dataset. This column will contain binary values representing the presence of cyberbullying: if the "cb_type" column indicates "not_cyberbullying," the corresponding "cb_label" entry will be set to 0, signifying that the text does not contain cyberbullying content.

```
# Add new column at the specific position


def cb_lbl(row):
    if row["cb_type"] == "not_cyberbullying":
        return 0
    else:
        return 1


# insert(location, columnName, value)
data.insert(1, "cb_label", data.apply(cb_lbl, axis=1))

data
```

✓ 0.2s

	text	cb_label	cb_type
0	In other words #katandandre, your food was cra...	0	not_cyberbullying
1	Why is #aussietv so white? #MKR #theblock #ImA...	0	not_cyberbullying
2	@XochitlSuckkks a classy whore? Or more red ve...	0	not_cyberbullying
3	@Jason_Gio meh. :P thanks for the heads up, b...	0	not_cyberbullying
4	@RudhoeEnglish This is an ISIS account pretend...	0	not_cyberbullying
...	...	...	...
47687	Black ppl aren't expected to do anything, depe...	1	ethnicity
47688	Turner did not withhold his disappointment. Tu...	1	ethnicity
47689	I swear to God. This dumb nigger bitch. I have...	1	ethnicity
47690	Yea fuck you RT @therealexel: IF YOURE A NIGGE...	1	ethnicity
47691	Bro. U gotta chill RT @CHILLShrammy: Dog FUCK ...	1	ethnicity

47692 rows × 3 columns

Figure 21: Source code and output of adding new column.

Merge the aggression_parsed_dataset and attack_parsed_dataset, ensuring to eliminate any duplicate entries. Introduce a new column to this combined dataset, designated for indicating the type of cyberbullying. In cases where the text is identified as involving cyberbullying, label this new column with the term “aggression”.

```
# Merge two datasets together
data1 = pd.concat([data1, data2], ignore_index=True)

# Remove duplicates (data in entire row is totally same)
data1 = data1.drop_duplicates()

# Add new cb_type column


def cb_type(row):
    if row["cb_label"] == 0:
        return "not_cyberbullying"
    else:
        return "aggression"


# insert(location, columnName, value)
data1.insert(2, "cb_type", data1.apply(cb_type, axis=1), allow_duplicates=True)

data1.sample(8)
```

✓ 1.2s

	text	cb_label	cb_type
62372	` == Talkback == `	0	not_cyberbullying
39160	I only want people to stop concerning my IP ...	0	not_cyberbullying
53450	::It was Meet the Quagmires, the episode wher...	0	not_cyberbullying
114843	FUCK YOU NEIIN! ACCEPT IT! IT DOESN'T CARE IF...	1	aggression
106121	. Ok Dennis Brown thanks for telling me I was ...	1	aggression
76549	` :::Best wishes, Fastily. May your future be ...	0	not_cyberbullying
72793	:This topic has started to degrade into perso...	0	not_cyberbullying
75760	:::But I can comment on their ACTS, I can co...	0	not_cyberbullying

Figure 22: Source code and output of joining aggression_parsed_dataset and attack_parsed_dataset.

d. Merge Datasets

Merge all data selected into one dataset using “concat” function in pandas library.

```
merged_df = pd.concat([data, data1, data2, data4, data5, data8])
```

Figure 23: Source code of merging datasets.

	text	cb_label	cb_type
58170	`== Re:List of wind farm projects in Romania...	0	not_cyberbullying
8445	#KellyFile #Hannity #TCOT	1	gender
4064	@harmlesstree2 The first Crusades were by the ...	1	racism
53468	`==please observe these Wiki guidelines on th...	0	not_cyberbullying
54181	[snappycomputer.com computer repair memphis]...	0	not_cyberbullying
53259	`:You have been asked by two editors both on...	0	not_cyberbullying
16685	@BilalIGhumman @OdiniaInvictus @IsraeliRegime ...	1	religion
63722	== LEAVE MY FUCKING PAGE ALONE! == LEAVE MY...	1	aggression
20541	, 2 November 2006 (UTC) I think it's getting ...	0	NaN
4808	== Please help with a dispute == User Imdak...	0	not_cyberbullying
4234	You need to look up the difference between a ...	0	not_cyberbullying
23248	EDIT-woops, that was me.	0	not_cyberbullying
23624	WHY are so many of you idiots getting Ganesha ...	1	religion
4674	This show is officially fucked!!!!! Space Cade...	0	not_cyberbullying
21988	But ur not deaf! I also have the full tamar Br...	0	not_cyberbullying

Figure 24: The dataset collected initially.

e. Prepare two datasets for cyberbullying detection and cyberbullying annotation

Prepare dataset for cyberbullying detection by removing the type columns from dataset merged in previous step. Then, remove the duplicates in the new dataset. The new dataset, df_detection consists only two columns: tweet text and label (1 or 0).

```
# for cyberbullying detection
df_detection = merged_df.drop("cb_type", axis=1)

df_detection.info()
```

✓ 0.0s

```
<class 'pandas.core.frame.DataFrame'>
Index: 461126 entries, 0 to 3463
Data columns (total 2 columns):
 #   Column    Non-Null Count   Dtype
---  ------    --------------   -----
 0   text      461126 non-null  object
 1   cb_label  461126 non-null  int64
dtypes: int64(1), object(1)
memory usage: 10.6+ MB
```

```
# remove duplicates
df_detection = df_detection.drop_duplicates("text")
df_detection.info()
```

✓ 0.2s

```
<class 'pandas.core.frame.DataFrame'>
Index: 251992 entries, 0 to 3463
Data columns (total 2 columns):
 #   Column    Non-Null Count   Dtype
---  ------    --------------   -----
 0   text      251992 non-null  object
 1   cb_label  251992 non-null  int64
dtypes: int64(1), object(1)
memory usage: 5.8+ MB
```

Figure 25: Source code of preparing dataset for cyberbullying detection.

	text	cb_label
81347	: As this has been here for some time with po...	0
41876	Watchin deep bluestupid dumb white bitxch Who ...	1
32052	He shot up a school, fuck the dude. He deserve...	1
98682	The five pillars of Wikipedia *Tutorial *How t...	0
20203	` Gleason, Philip. ``Trouble in the Colonial ...	0
20746	Your account or IP address has been blocked ...	0
99824	No I do not appreciate this. I have sold data...	0
102604	`:List of micronations is intended to be a l...	0
38147	:That's exactly why I wrote about complete no...	0
826	that will substantiate your claim that the Ir...	0

Figure 26: The dataset collected for cyberbullying detection.

From the merged dataset, select only those records that have been identified as containing cyberbullying. These selected records will then be utilized for the development of a cyberbullying annotation system.

```
# take only the records that detected cyberbullying

# remove the records that are not cyberbullying
df_annotation = merged_df[merged_df['cb_label'] != 0]

# remove 'cb_label' column
df_annotation = df_annotation.drop("cb_label", axis=1)
df_annotation.info()
```

✓ 0.0s

```
<class 'pandas.core.frame.DataFrame'>
Index: 89587 entries, 7945 to 3459
Data columns (total 2 columns):
 #   Column   Non-Null Count  Dtype
---  ------   --------------  -----
 0   text     89587 non-null  object
 1   cb_type  75997 non-null  object
dtypes: object(2)
memory usage: 2.1+ MB
```

```
# remove duplicates
df_annotation = df_annotation.drop_duplicates("text")
df_annotation.info()
```

✓ 0.0s

```
<class 'pandas.core.frame.DataFrame'>
Index: 60414 entries, 7945 to 3459
Data columns (total 2 columns):
 #   Column   Non-Null Count  Dtype
---  ------   --------------  -----
 0   text     60414 non-null  object
 1   cb_type  60414 non-null  object
dtypes: object(2)
memory usage: 1.4+ MB
```

Figure 27: Source code of preparing dataset for cyberbullying annotation.

	text	cb_type
114821	== Re: BOI bias == Not at all man. No bias ...	aggression
8371	Cancel culture is a fair thing to use when som...	gender
43209	Laundry teaches kids racism. Separate the blac...	ethnicity
14993	Ha. It's funny. Get it. Because men are raping...	gender
7852	I also find it interesting that after TXmast...	aggression
19438	You'd rather wish all the Muslim women sit at ...	religion
17745	I don't need to debate with idiots like you! I...	religion
14576	My mother in law is like that about her dog. I...	gender
108876	:: DNA analysis confirms that the 2 individu...	toxicity
46294	@tayyoung_: FUCK OBAMA, dumb ass nigger -- lol...	ethnicity

Figure 28: The dataset collected for cyberbullying annotation.

3.5 Initial Data Pre-processing

During this stage, the text data in the datasets is necessary to perform data cleaning before proceeding to the feature extraction/further steps. Figure 29 shows the function of data cleaning to be run on each tweets data.

```
def clean_text(text):
    # remove non-alphabetic characters and convert to lowercase
    text = re.sub("[^a-zA-Z]", " ", text).lower()

    # tokenize the text
    words = nltk.word_tokenize(text)

    # remove stopwords
    words = [w for w in words if w not in stopwords.words("english")]

    # word stemming
    ps = PorterStemmer()
    words = [ps.stem(w) for w in words]

    # Join the words back into a string
    text = " ".join(words)

    return text
```

Figure 29: Data cleaning function.

```
df_detection.head()
✓ 0.0s
```

	text	cb_label
0	In other words #katandandre, your food was cra...	0
1	Why is #aussietv so white? #MKR #theblock #ImA...	0
2	@XochitlSuckkks a classy whore? Or more red ve...	0
3	@Jason_Gio meh. :P thanks for the heads up, b...	0
4	@RudhoeEnglish This is an ISIS account pretend...	0

```
# perform data cleaning on df_detection
# df_detection["text"] = df_detection["text"].apply(clean_text)

# test cleaning first 10 rows of data
df_detection.iloc[0:10, df_detection.columns.get_loc("text")] = df_detection.iloc[0:10]["text"].apply(clean_text)

df_detection.head()
✓ 0.1s
```

	text	cb_label
0	word katandandr food crapilici mkr	0
1	aussietv white mkr theblock imacelebrityau tod...	0
2	xochitlsuckkk classi whore red velvet cupcak	0
3	jason gio meh p thank head concern anoth angri...	0
4	rudhoeenglish isi account pretend kurdish acco...	0

Figure 30: Comparison result before and after performed data cleaning.

3.5.1 Remove punctuation and convert to lowercase

Remove punctuation and other non-letter characters using regular expressions and convert all letters to lowercase.

3.5.2 Word Tokenization

Word tokenization is the process of splitting a piece of text into individual words, or "tokens". In Natural Language Processing (NLP), this is a fundamental step as it allows for the analysis of word usage and frequency, among other things. The Natural Language Toolkit (NLTK) is a popular Python library that provides a suite of text processing libraries for classification, tokenization, stemming, tagging, parsing, and semantic reasoning, among other NLP tasks.

3.5.3 Remove Stop Words

Stopwords are commonly used words in any language (like "the", "is", "in", "for") that are often filtered out before processing text because they carry very little useful information about the content of the text. NLTK provides a corpus of stopwords that can be used to filter stopwords from text.

3.5.4 Word Stemming

Word stemming is a process in Natural Language Processing (NLP) aimed at reducing words to their word stem, base or root form - generally a written word form. The PorterStemmer is a popular stemming algorithm that is widely used for this purpose.

3.6 Summary

Chapter 3 of the thesis outlines the methodology adopted for the data analysis project, focusing on the CRISP-DM framework. This framework is chosen for its structured, step-by-step approach, which is beneficial for guiding the project from understanding the business problem to deploying the solution. The methodology is cyclical, allowing for continuous improvement and adaptation to new insights. The chapter details the six phases of CRISP-DM:

Business Understanding, Data Understanding, Data Preparation, Modelling, Evaluation, and Deployment, each with specific objectives and tasks.

Data collection is primarily from Kaggle.com, encompassing various datasets related to cyberbullying. The Data Understanding phase involves loading, exploring, and comparing these datasets to ascertain their relevance and applicability. This phase includes detailed analysis and comparison of datasets to identify overlaps and unique characteristics, using tools like datacompy for comparison.

The initial data pre-processing steps are outlined, focusing on selecting relevant data, renaming columns for consistency, and merging datasets. Two specific datasets are prepared: one for cyberbullying detection and another for cyberbullying annotation. The chapter also describes the initial data pre-processing steps, including data cleaning, punctuation removal, word tokenization, stop word removal, and word stemming, using tools like NLTK. These steps are crucial for preparing the text data for feature extraction and further analysis in the subsequent part of the project.

CHAPTER 4: CONCLUSION

4.1 The achievement of the first part of the project

In conclusion, this project marks a significant achievement in the realm of cyberbullying research. The researcher successfully prepared two distinct datasets, each tailored for a specific facet of the study. The first dataset, aimed at cyberbullying detection, was meticulously compiled to identify instances of cyberbullying within digital communications. The second dataset, dedicated to cyberbullying annotation, served the purpose of categorizing these instances into specific types of cyberbullying, based on various criteria such as age, gender, or other relevant factors.

Crucially, the researcher not only gathered these datasets but also effectively cleaned and processed them. This cleaning process involved removing irrelevant data, filtering out noise, and standardizing the datasets for analysis, which is a critical step in ensuring the accuracy and reliability of any machine learning or data analysis task. By doing so, the datasets were transformed into valuable resources for developing and training models that can accurately detect and classify instances of cyberbullying.

This accomplishment sets a solid foundation for the next stages of the project, which involve applying machine learning techniques to analyse the data. The prepared datasets are instrumental in training models to understand and predict cyberbullying patterns, potentially leading to effective tools for monitoring and mitigating cyberbullying in digital spaces. The success of this initial phase is a testament to the meticulous and thoughtful approach taken by the researcher, and it paves the way for meaningful contributions to the field of cyberbullying research and prevention.

4.2 Justify the investigation or research for the project

The researcher's efforts in this project demonstrate a thorough and comprehensive approach to addressing the challenge of cyberbullying through data-driven analysis. The successful preparation and cleaning of two specialized datasets—one for the detection of

cyberbullying and another for its annotation—indicate a deep level of investigation and research that aligns well with the project's objectives.

The exploration of feature extraction techniques reflects a keen understanding of how crucial this step is in preparing the data for effective model training. By delving into various methods of transforming raw text data into usable features, the researcher has ensured that the foundation for the machine learning models is sound and reliable. This demonstrates not just a theoretical grasp of the concepts but also a practical application of them.

Moreover, the researcher's investigation into machine learning algorithms showcases an adeptness in identifying the most suitable models for the project's goals. This involves understanding the nuances of each algorithm, their strengths and weaknesses, and how they can be best applied to the specific context of cyberbullying detection and annotation. The selection of algorithms, therefore, is not just based on their popularity or general efficacy, but is a decision informed by comprehensive research and tailored to the project's unique needs.

4.3 Research and areas that need further explore and improvement

There remains an area for further exploration and improvement: the deployment of a web application that integrates the developed model, as outlined in the project objectives. The gap in the current research primarily lies in the practical application and deployment of the machine learning model in a real-world context. The process of deploying a model on a website involves several critical steps beyond model development, including:

1. **Web Development Skills**: Building a user-friendly and functional website requires knowledge in web development, including front-end and back-end programming. Familiarity with web frameworks, such as Flask or Django for Python, would be beneficial. The student may need to either develop these skills or collaborate with web developers.
2. **Model Integration**: Integrating the machine learning model into a web application poses its own set of challenges. This includes setting up an API for the model, ensuring

the model runs efficiently in a server environment, and handling real-time requests and responses.

3. **Testing and Maintenance**: Continuous testing is essential to ensure the website and model perform as intended, along with regular updates and maintenance post-deployment.

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Appendices

Appendix A: PPF – Title Registration Proposal

Office Record	Receipt
Date Received: Received by whom:	Student name: Student number: Received by: Date:

DRAFT PROJECT PROPOSAL FORM

Proposal ID :

Supervisor :

1) **Mr. Raheem Mafas**
2) **Dr. Dewi Octaviani**
3) **Dr. Preethi Subramanian**
4) **Ms. Minnu Helen Joseph**
5) **Mr. Mohammad Namazee**

Student Name : TEH KAI LING

Student No. : TP063890

Email Address : tp063890@mail.apu.edu.my & kailingteh@hotmail.com

Programme Name : Bachelor in Computer Science (Data Analytics) (CSDA)

Title of project : Enhancing Online Safety: A Text Analysis Approach For Cyberbullying Detection In Social Media Tweets

Please record which module(s) your topic is related to:

Text Analytics and Sentiment Analysis (CT107-3-3-TXSA)
Data Mining and Predictive Modelling (CT119-3-2-DMPM)
Data Management (CT075-3-2-DTM)

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1. Introduction

Assume the reader has very little knowledge of the subject. Introduce the topic, the sector of business/industry concerned and how the project relates to it. Define the context of the problem and identify the research required to solve it.

The widespread use of social media platforms has completely transformed the way we connect, communicate, and share information in the quickly changing digital world. These platforms have brought people from all corners of the globe closer together, fostering a sense of global interconnectedness like never before. However, this digital transformation has also given rise to a darker side, one that is characterised by the advent of cyberbullying, a widespread and upsetting problem that affects people's safety and well-being online. (Chan et al., 2021)

Cyberbullying detection in social media tweets is a crucial topic that needs to be addressed in the broader context of online safety and social media management in order to bring awareness to a problem that has significant ramifications for people, businesses, and society at large. (Mehendale et al., 2022) The need to address cyberbullying and harassment has increased as more people and companies rely on social media platforms to interact with others. (Yi & Zubiaga, 2023) Cyberbullying can take many different forms, such as distributing harmful content, making harsh comments, and making personal assaults on the victim. All these forms of cyberbullying can have serious emotional and psychological repercussions for its targets. (Garett et al., 2016) Thus, it has become imperative for social media platforms and online communities to take proactive measures to detect and combat cyberbullying effectively.

This project explores the relationship between technology, social responsibility, and digital well-being. It seeks to address the pressing issue of cyberbullying by harnessing the power of text analysis and natural language processing (NLP) techniques. It intends to contribute to the continuous efforts to create safer digital environments for people.

2. Problem Statement

Identify past and current work in the subject area. Outline the key references to other people's work, indicate for the most pertinent of these how your proposal relates to the ideas they contain.

1. Rising Incidence of Cyberbullying

Cyberbullying is still prevalent on social media platforms today, harming people and negatively affecting how people use the internet (Hinduja & Patchin, 2018). With the increasing adoption of social media, the prevalence of cyberbullying occurrences has rapidly increased, posing a significant threat to online safety and the emotional well-being of users. As this problem worsens, there is an urgent need for the creation of efficient tools and tactics for spotting and preventing cyberbullying in social network tweets. Addressing this issue is crucial for preventing problems like depression, suicidal thoughts, and other mental health difficulties that can arise from the unpleasant experience of online harassment in addition to improving online safety.

2. Ineffective Manual Moderation

The precision and effectiveness needed for real-time identification and mitigation of online harassment inside social media platforms are frequently not met by current research and solutions in cyberbullying detection. This inadequacy puts people at risk of damage and emphasises the urgent need for the creation of more sophisticated and accurate detection tools. The current approaches' reliance on manual moderation is untenable due to its time-consuming and costly nature, especially considering the overwhelming volume of content on social media. Automating the detection and reaction to cyberbullying episodes while reducing the inherent problems of false positives and negatives is clearly necessary to address these concerns (Wu et al., 2020).

3. Complexity of Natural Language

Detecting cyberbullying is challenging due to the ever-changing strategies used by cyberbullies, making it impossible to discern their damaging purpose within the complexities of natural language. The prevalence of online harassment is continuously changing, making it more difficult to develop measures for detection and prevention that are effective (Riebel et al., 2019). Furthermore, while most of the cyberbullying research focuses on English-language content, social media's global spread needs the creation of models capable of detecting cyberbullying across multiple languages and cultural contexts (Ptaszynski et al., 2020).

3. Project Aim and Objectives

Identify the AIM(s) of the project, i.e. what the overall achievement is intended to be, in terms of both academic and commercial/industrial advances. Identify the particular intellectual difficulties posed by the proposal, the problems to be addressed, and explain how these might be solved. Clearly list individual measurable OBJECTIVES which can be related to the workplan and deliverables.

Aims and objectives are subject to approval from supervisor and students are expected to revise them if deemed inappropriate for a Level 3 project.

The aim of this project is to develop an automated cyberbullying detection system that utilizes text analysis techniques to identify and mitigate cyberbullying in social media tweets effectively.

The objectives of this project are:

1. To develop an automated cyberbullying detection system that integrates text analysis techniques to identify cyberbullying in social media tweets.
2. To categorise the detected cyberbullying content into different types, including hate speech, harassment, and toxicity, for precise identification and response.
3. To implement effective text analysis and natural language processing methods to enhance the accuracy and efficiency of the cyberbullying detection system.
4. To enable real-time cyberbullying detection within social media platforms to ensure that potentially harmful content is identified and addressed promptly.

4. Literature Review

In the field of cyberbullying detection, Balakrishnan et al. (2020) published a significant study in the publication titled "Improving Cyberbullying Detection Using Twitter Users' Psychological Features and Machine Learning." An in-depth analysis of the complex relationship between Twitter users' psychological characteristics and their propensity to engage in cyberbullying activities is provided in this study. The study aims to identify associations between personality qualities from the Big Five model (BFM) and propensities for cyberbullying. The researchers offer substantial empirical support for their contention that certain psychological characteristics, such as particular personality traits, and online crimes like cyberbullying, are strongly related. The work gives a new viewpoint on improving cyberbullying detection systems by considering the psychological profiles of users by comprehending these relationships.

The study emphasises the critical role that textual characteristics have in identifying cyberbullying on a variety of social media sites. (Fortunatus et al., 2020) The study proposes a novel rule-based method called the lexicon-enhanced rule-based method, which combines conventional rule-based methods with an extensive lexicon designed specifically for the identification of cyberbullying. By combining the strength of rule-based algorithms and specialised lexicons, this approach aims to increase the detection's precision and effectiveness.

The findings of this study highlight the possibility of fusing different textual elements to build a reliable and efficient cyberbullying detection system.

The research stands out as groundbreaking work in the constantly changing fields of machine learning and natural language processing. (Muneer et al., 2023) This study offers a novel ensemble stacking learning strategy designed specifically for the identification of Twitter cyberbullying actions. The study intends to considerably improve the accuracy and reliability of cyberbullying detection by utilising Deep Neural Network methods (DNNs) and incorporating them with the sophisticated BERT model. The ensemble approach ensures a thorough examination of tweets by combining the advantages of various machine learning models. The BERT model's incorporation improves the system's capacity to comprehend the subtleties and complexities of real language, improving the accuracy and efficacy of the detection process.

5. Deliverables

Provide a clear list of the outputs from the project.

The following are the steps that will be implemented in this project to develop an effective cyberbullying detection system:

1. Data Collection

The data collection for this project involved gathering a diverse dataset comprising information from various sources related to the automatic detection of cyberbullying. These sources included popular social media platforms such as Kaggle, Twitter, Wikipedia Talk pages, and YouTube. The dataset consisted of text data, each labelled as either indicative of cyberbullying or not. This multi-source dataset was essential for training and evaluating effective models for cyberbullying detection across various online contexts and forms of abusive behaviour.

2. Data Preprocessing & Data Understanding

Understanding the unstructured text data can be an iterative and exploratory process. Data preprocessing text data includes removing irrelevant symbols or emojis, word or sentence

tokenizing, stemming the words to the base form and even normalization such as lowercasing all words to ensure consistency. Then, get the basic statistics such as the number of tweets or comments, average tweet length, and word frequency distribution which show an overview of the dataset. Furthermore, using syntactic analysis tools or libraries and applying topic modelling techniques will generally give a better understanding of the content in the comments or tweets.

3. Model Training

The dataset is split into training, validation and testing sets in this phase. The training data is used to train the model, validation data helps to refine the model and testing data is used to evaluate its performance. The appropriate method will be used to train the model. For example, Gaussian Naïve Bayes, Decision Tree, K Nearest Neighbors, Random Forest, SVM (Support Vector Machine), Logistic Regression and custom LSTM w/ attention RNN by PyTorch, BERT.

4. Model Evaluation

Evaluating the model is crucial to assess its effectiveness. The common method to evaluate the model is understanding the strengths and weaknesses of the model using a confusion matrix. (Desai et al., 2021) This matrix provides a clear breakdown of the model's predictions, revealing the number of true positives (instances correctly identified as cyberbullying), true negatives (non-cyberbullying instances correctly identified), false positives (non-cyberbullying instances incorrectly classified as cyberbullying), and false negatives (cyberbullying instances missed by the model). Another method that can be used to evaluate the model is the F1 score which measures the accuracy of the model trained. (Evangeline et al., 2021) There are other ways to evaluate the model such as generating performance metrics using Lazy Predict, Precision-Recall Curve or ROC Curve generation.

5. Model Deployment

The cyberbullying detection model will be presented on a simple website. Users can input words, phrases, or sentences, and the model will promptly assess whether the content contains cyberbullying (binary classification, either cyberbullying or not-cyberbullying). If cyberbullying is detected, the model will also provide an analysis categorizing it into specific cyberbullying categories for a more detailed understanding of the content's nature (qualitative classification).

The cyberbullying detection model can be valuable for various user groups and organizations concerned about online safety and well-being. Any organization such as social media platforms, educational institutions, online communities or regulatory bodies can use the cyberbullying detection model to automatically identify and mitigate cyberbullying incidents on their platforms, ensuring a safer online environment for their users. Upon using the model, organizations have the flexibility to determine the actions to be taken upon detecting cyberbullying comments or tweets, whether it involves warning the user, removing the content, or notifying relevant authorities, allowing for tailored responses that align with their specific safety policies and guidelines.

6. References

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Appendix B: Ethics Forms (Fast Track)

Office Record	Receipt – Fast-Track Ethical Approval
Date Received:	Student name: Student number:
Received by whom:	Received by: Date:

APU / APIIT FAST-TRACK ETHICAL APPROVAL FORM (STUDENTS)

Tick one box (level of study):
- ☐ POSTGRADUATE (PhD / MPhil / Masters)
- √ **UNDERGRADUATE (Bachelors degree)**
- ☐ FOUNDATION / DIPLOMA / Other categories

Tick one box (purpose of approval):
- √ ~~Thesis / Dissertation~~ / **FYP project**
- ☐ Module assignment
- ☐ Other: _____

Title of Programme on which enrolled **Bachelor in Computer Science (Data Analytics)**

Tick one box: √ **Full-Time Study** or ☐ Part-Time Study

Title of project / assignment **Enhancing Online Safety: A Text Analysis Approach For Cyberbullying Detection In Social Media Tweets**

Name of student researcher **Teh Kai Ling**

Name of supervisor / lecturer **Dr. Dewi Octaviani**

Student Researchers- please note that certain professional organisations have ethical guidelines that you may need to consult when completing this form.

Supervisors/Module Lecturers - please seek guidance from the Chair of the APU Research Ethics Committee if you are uncertain about any ethical issue arising from this application.

		YES	NO	N/A
1	Will you describe the main procedures to participants in advance, so that they are informed about what to expect?			√
2	Will you tell participants that their participation is voluntary?			√
3	Will you obtain written consent for participation?			√
4	If the research is observational, will you ask participants for their consent to being observed?			√
5	Will you tell participants that they may withdraw from the research at any time and for any reason?			√
6	With questionnaires and interviews will you give participants the option of omitting questions they do not want to answer?			√
7	Will you tell participants that their data will be treated with full confidentiality and that, if published, it will not be identifiable as theirs?			√
8	Will you give participants the opportunity to be debriefed i.e. to find out more about the study and its results?			√

If you have ticked **No** to any of Q1-8 you should complete the full Ethics Approval Form.

		YES	NO	N/A
9	Will your project/assignment deliberately mislead participants in any way?			√
10	Is there any realistic risk of any participants experiencing either physical or psychological distress or discomfort?			√
11	Is the nature of the research such that contentious or sensitive issues might be involved?			√

If you have ticked **Yes** to 9, 10 or 11 you should complete the full Ethics Approval Form. In relation to question 10 this should include details of what you will tell participants to do if they should experience any problems (e.g. who they can contact for help). You may also need to consider risk assessment issues.

Fast-Track Ethical Approval Form ver. 3.0 (Dec 2015) Page 1 of 4

			YES	NO	N/A
12	Does your project/assignment involve work with animals?			√	
13	Do participants fall into any of the following special groups? **Note that you may also need to obtain satisfactory clearance from the relevant authorities**	Children (under 18 years of age)		√	
		People with communication or learning difficulties			
		Patients			
		People in custody			
		People who could be regarded as vulnerable			
		People engaged in illegal activities (eg drug taking)			
14	Does the project/assignment involve external funding or external collaboration where the funding body or external collaborative partner requires the University to provide evidence that the project/assignment had been subject to ethical scrutiny?			√	

If you have ticked **Yes** to 12, 13 or 14 you should complete the full Ethics Approval Form. There is an obligation on student and supervisor to bring to the attention of the APU Research Ethics Committee any issues with ethical implications not clearly covered by the above checklist.

STUDENT RESEARCHER
Provide in the boxes below (plus any other appended details) information required in support of your application, THEN SIGN THE FORM.

Please Tick Boxes

I consider that this project/assignment has no significant ethical implications requiring a full ethics submission to the APU Research Ethics Committee.	√
Give a brief description of participants and procedure (methods, tests used etc) in up to 150 words. This project is focused on detecting and categorizing cyberbullying content within social media tweets through advanced text analysis methods. This project will use the datasets from Kaggle.com. There are nine datasets that encapsulate the tweets or comments from various social media platforms, including Twitter and YouTube. Some of these datasets provide labels indicating the category of cyberbullying. The links to the dataset are below: https://www.kaggle.com/datasets/saurabhshahane/cyberbullying-dataset/data https://www.kaggle.com/datasets/andrewmvd/cyberbullying-classification This project will apply CRISP-DM as the data analysis methodology. Only essential columns will be selected for further process. Subsequent data preprocessing steps will include tokenization, conversion to lowercase, removal of stop words, normalization, and lemmatization. Necessary data visualizations will be undertaken throughout the project. Naïve Bayes, Regression Model, Decision Tree and RNN will be utilized for model training. The models will then be evaluated using a confusion matrix and F1 score. There is no survey or interview to be conducted for this project.	
I also confirm that: i) All key documents e.g. consent form, information sheet, questionnaire/interview are appended to this application.	N/A

Or ii) Any key documents e.g. consent form, information sheet, questionnaire/interview schedules which need to be finalised following initial investigations will be submitted for approval by the project/assignment supervisor/module lecturer before they are used in primary data collection.	

E-signature *Teh Kai Ling* Print Name Teh Kai Ling Date 5/10/2023
(Student Researcher)

Please note that any variation to that contained within this document that in any way affects ethical issues of the stated research requires the appending of new ethical details. New ethical consent may need to be sought.
The completed form (and any attachments) should be submitted for consideration by your Supervisor/Module Lecturer

SUPERVISOR/MODULE LECTURER
PLEASE CONFIRM THE FOLLOWING:

	Please Tick Box
I consider that this project/assignment has no significant ethical implications requiring a full ethics submission to the APU Research Ethics Committee	√
i) I have checked and approved the key documents required for this proposal (e.g. consent form, information sheet, questionnaire, interview schedule) Or	N/A
ii) I have checked and approved draft documents required for this proposal which provide a basis for the preliminary investigations which will inform the main research study. I have informed the student researcher that finalised and additional documents (e.g. consent form, information sheet, questionnaire, interview schedule) must be submitted for approval by me before they are used for primary data collection.	

SUPERVISOR AND SECOND ACADEMIC SIGNATORY

STATEMENT OF ETHICAL APPROVAL (please delete as appropriate)

1) **THIS PROJECT/ASSIGNMENT HAS BEEN CONSIDERED USING AGREED APU/SU PROCEDURES AND IS NOW APPROVED**

2) **THIS PROJECT/ASSIGNMENT HAS BEEN APPROVED IN PRINCIPLE AS INVOLVING NO SIGNIFICANT ETHICAL IMPLICATIONS, BUT FINAL APPROVAL FOR DATA COLLECTION IS SUBJECT TO THE SUBMISSION OF KEY DOCUMENTS FOR APPROVAL BY SUPERVISOR (see Appendix A)**

E-signature *Dewi* Print Name Dr. Dewi Octaviani Date 24/10/2023
(Supervisor/Lecturer)

E-signature… … … … … … … Print Name Ms. Farhana Illiani Binti Hassan Date… … … …
(Second Academic Signatory)

Office Record	Receipt – Appendix A (Fast-Track Ethics Form)
Date Received:	Student name: Student number: Received by:
Received by whom:	Date:

APPENDIX A
AUTHORISATION FOR USE OF KEY DOCUMENTS

Completion of Appendix A is required when for good reasons key documents are not available when a fast track application is approved by the supervisor/module lecturer and second academic signatory.

I have now checked and approved all the key documents associated with this proposal e.g. consent form, information sheet, questionnaire, interview schedule

Title of project/assignment… … … …

Appendix C: Log Sheets

Log Sheet 1

(APU: Serial Number)

PLS V1.0

Project Log Sheet – Supervisory Session

Notes on use of the project log sheet:

1. This log sheet is designed for meetings of more than 15 minutes duration, of which there must be at minimum SIX (6) during the course of the project (SIX mandatory supervisory sessions).
2. The student should prepare for the supervisory sessions by deciding which question(s) he or she needs to ask the supervisor and what progress has been made (if any) since the last session, and noting these in the relevant sections of the form, effectively forming an agenda for the session.
3. A log sheet is to be brought by the STUDENT to each supervisory session.
4. The actions by the student (and, perhaps the supervisor), which should be carried out before the next session should be noted briefly in the relevant section of the form.
5. The student should leave a copy (after the session) of the Project Log Sheet with the supervisor and to the administrator at the academic counter. A copy is retained by the student to be filed in the project file.
6. It is recommended that students bring along log sheets of previous meetings together with the project file during each supervisory session.
7. The log sheet is an important deliverable for the project and an important record of a student's organisation and learning experience. The student **must** hand in the log sheets as an appendix of the final year documentation, with sheets dated and numbered consecutively.

Student's name: Teh Kai Ling **Date:** 18/10/2023 **Meeting No:** 1

Project title: Enhancing Online Safety: A Text Analysis Approach For Cyberbullying Detection In Social Media Tweets **Intake:** APU3F2308CS(DA)

Supervisor's name: Dr. Dewi Octaviani **Supervisor's signature:** *Dewi*

Items for discussion (noted by student before mandatory supervisory meeting):

1. **Confirm the datasets selected.**
2. **Review Ethics Form (Fast Track).**
3. **Update IR progression.**
4. **Ask questions about the datasets selected and IR.**

Record of discussion (noted by student during mandatory supervisory meeting):

1. **Understand my deliverables / what I planned to do in this project.**
2. **Correct the objectives, give a clear direction to this project.**
3. **Go through each section of IR, discuss the idea and what should be written in the section.**
4. **Select SEMMA as methodology is not suitable, changed to CRISP-DM.**
5. **Need to find the best and optimize way to handle the datasets. E.g., fill missing values, remove column, …**

Action List (to be attempted or completed by student by the next mandatory supervisory meeting):

1. **Send the ethic form and project log sheet to supervisor to sign.**
2. **Next meeting will be after 3 weeks.**
3. **Complete the first draft of IR before next meeting.**

Note: A student should make an appointment to meet his or her supervisor (via the consultation system) at least ONE (1) week prior to a mandatory supervisor session – please see document on project timelines. In the event a supervisor could not be booked for consultation, the project manager should be informed ONE (1) week prior to the session so that a meeting can be subsequently arranged.

Project Log Sheet

Log Sheet 2

(APU: Serial Number)

PLS V1.0

Project Log Sheet – Supervisory Session

Notes on use of the project log sheet:

1. This log sheet is designed for meetings of more than 15 minutes duration, of which there must be at minimum SIX (6) during the course of the project (SIX mandatory supervisory sessions).
2. The student should prepare for the supervisory sessions by deciding which question(s) he or she needs to ask the supervisor and what progress has been made (if any) since the last session, and noting these in the relevant sections of the form, effectively forming an agenda for the session.
3. A log sheet is to be brought by the STUDENT to each supervisory session.
4. The actions by the student (and, perhaps the supervisor), which should be carried out before the next session should be noted briefly in the relevant section of the form.
5. The student should leave a copy (after the session) of the Project Log Sheet with the supervisor and to the administrator at the academic counter. A copy is retained by the student to be filed in the project file.
6. It is recommended that students bring along log sheets of previous meetings together with the project file during each supervisory session.
7. The log sheet is an important deliverable for the project and an important record of a student's organisation and learning experience. The student **must** hand in the log sheets as an appendix of the final year documentation, with sheets dated and numbered consecutively.

Student's name: Teh Kai Ling **Date:** 8/11/2023 **Meeting No: 2**

Project title: Enhancing Online Safety: A Text Analysis Approach For Cyberbullying Detection In Social Media Tweets **Intake:** APU3F2308CS(DA)

Supervisor's name: Dr. Dewi Octaviani **Supervisor's signature:** *Dr. Dewi*

Items for discussion (noted by student before mandatory supervisory meeting):

1. **Let supervisor to review IR draft.**

Record of discussion (noted by student during mandatory supervisory meeting):

Feedback from supervisor after reviewing the draft:

1. Many important components are missing.
2. Chapter 1: Include more citations to support the project's background and problem statements.
3. Chapter 2: Provide a more detailed review of the research domain to meet the expected academic or industry standards.
4. Chapter 3: Explain data understanding and preprocessing in more detail.
5. Use more references from reputable papers throughout the document to strengthen your arguments and show your familiarity with existing research.

Action List (to be attempted or completed by student by the next mandatory supervisory meeting):

1. **Complete and enhance the investigation report based on the feedback from supervisor.**

Note: A student should make an appointment to meet his or her supervisor (via the consultation system) at least ONE (1) week prior to a mandatory supervisor session – please see document on project timelines. In the event a supervisor could not be booked for consultation, the project manager should be informed ONE (1) week prior to the session so that a meeting can be subsequently arranged.

Project Log Sheet

Log Sheet 3

(APU: Serial Number)

PLS V1.0

Project Log Sheet – Supervisory Session

Notes on use of the project log sheet:

1. This log sheet is designed for meetings of more than 15 minutes duration, of which there must be at minimum SIX (6) during the course of the project (SIX mandatory supervisory sessions).
2. The student should prepare for the supervisory sessions by deciding which question(s) he or she needs to ask the supervisor and what progress has been made (if any) since the last session, and noting these in the relevant sections of the form, effectively forming an agenda for the session.
3. A log sheet is to be brought by the STUDENT to each supervisory session.
4. The actions by the student (and, perhaps the supervisor), which should be carried out before the next session should be noted briefly in the relevant section of the form.
5. The student should leave a copy (after the session) of the Project Log Sheet with the supervisor and to the administrator at the academic counter. A copy is retained by the student to be filed in the project file.
6. It is recommended that students bring along log sheets of previous meetings together with the project file during each supervisory session.
7. The log sheet is an important deliverable for the project and an important record of a student's organisation and learning experience. The student **must** hand in the log sheets as an appendix of the final year documentation, with sheets dated and numbered consecutively.

Student's name: Teh Kai Ling **Date:** 28/11/2023 **Meeting No: 3**

Project title: Enhancing Online Safety: A Text Analysis Approach For Cyberbullying Detection In Social Media Tweets **Intake:** APU3F2308CS(DA)

Supervisor's name: Dr. Dewi Octaviani **Supervisor's signature:** *Dr. Dewi*

Items for discussion (noted by student before mandatory supervisory meeting):

1. **Final check on the investigation report.**

Record of discussion (noted by student during mandatory supervisory meeting):

1. **The student sent the IR in last minutes which not given ample time for supervisor to review thoroughly. This is due to poor time management.**
2. **There are some yellow highlights in the IR file – need to remove them.**
3. **There are some empty pages – such as list of figures, list of tables, etc.**
4. **Chapter 2 – lack of literature review, could be deeper and wider.**
5. **Chapter 3 – could add more discussions on the results of data understanding and data pre-processing**
6. **Chapter 4 – too brief, and should start on a new page.**

Action List (to be attempted or completed by student by the next mandatory supervisory meeting):

1. **Better time management**
2. **Improve the IR**

Note: A student should make an appointment to meet his or her supervisor (via the consultation system) at least ONE (1) week prior to a mandatory supervisor session – please see document on project timelines. In the event a supervisor could not be booked for consultation, the project manager should be informed ONE (1) week prior to the session so that a meeting can be subsequently arranged.

Project Log Sheet

Appendix D: Gantt Chart

ID	Task Name	Duration
1	Project Proposal Form	10 days
2	**CHAPTER 1: INTRODUCTION**	**5 days**
3	1.1 Introduction	2 days
4	1.2 Problem Background	2 days
5	1.3 Project Aim	1 day
6	1.4 Objectives	1 day
7	1.5 Scope	2 days
8	1.6 Potential Benefit	1 day
9	1.6.1 Tangible Benefit	1 day
10	1.6.2 Intangible Benefit	1 day
11	1.6.3 Target User	1 day
12	1.7 Overview of IR	1 day
13	1.8 Project Plan	1 day
14	**CHAPTER 2: LITERATURE REVIEW**	**14 days**
15	2.1 Introduction	1 day
16	2.2 Domain Research	4 days
17	2.3 Similar Systems/Works	4 days
18	2.4 Technical Research	4 days
19	2.5 Summary	1 day
20	**CHAPTER 3: METHODOLOGY**	**14 days**
21	3.1 Introduction	1 day
22	3.2 Methodology	4 days
23	3.3 Data Collection and Data Understa	4 days
24	3.4 Initial Data Pre-processing	4 days
25	3.5 Summary	1 day
26	**CHAPTER 4: CONCLUSION**	**2 days**
27	**REFERENCES**	1 day
28	**APPENDICES**	1 day

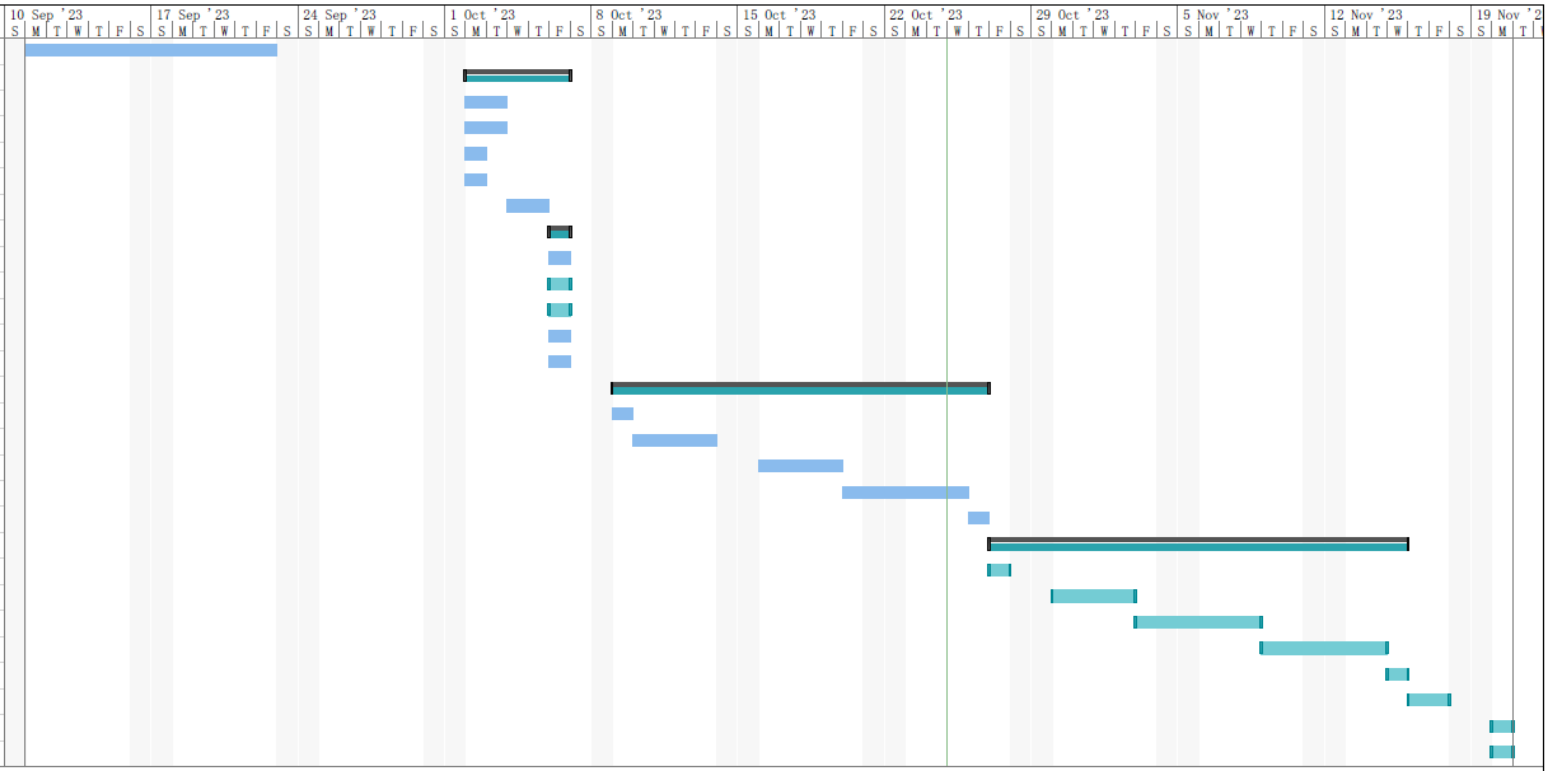


Figure 31: Gannt Chart